

Is the Distribution of Stock Returns Predictable?*

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Abstract

A large literature has considered predictability of the mean or volatility of stock returns but little is known about whether the distribution of stock returns more generally is predictable. We explore this issue in a quantile regression framework and consider whether a range of economic state variables are helpful in predicting different quantiles of stock returns representing left tails, right tails or shoulders of the return distribution. Many variables are found to have an asymmetric effect on the return distribution, affecting lower, central and upper quantiles very differently. Out-of-sample forecasts suggest that upper quantiles of the return distribution can be predicted by means of economic state variables although the center of the return distribution is more difficult to predict. Economic gains from utilizing information in time-varying quantile forecasts are demonstrated through portfolio selection and option trading experiments.

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1 Introduction

Risk averse investors generally require an estimate of the entire distribution of future stock returns to make their portfolio decisions. This holds under standard preferences such as constant relative risk aversion as well as under loss or disappointment aversion (Gul (1991)) or general preferences such as those considered by Kimball (1993). Empirical evidence confirms that investors' interest in stock returns goes well beyond their mean and variance. Studies such as Harvey and Siddique (2000) and Dittmar (2002) consider three and four-moment CAPM specifications and find that higher order moments help explain cross-sectional variation in US stock returns and have significant effects on expected returns.

In view of the economic importance of the full return distribution for asset pricing, risk management and asset allocation purposes, surprisingly little is known about which parts of the return distribution are predictable and how they depend on economic state variables. For example, is the probability of encountering a significant drop in stock prices time-varying? Are periods with surges in market prices predictable and linked to particular states of the economy? Answers to these questions have important portfolio implications and help improve our understanding of the economic sources of return predictability.

This paper proposes a novel approach to analyzing the predictability of different parts of the distribution of stock returns as represented by its individual quantiles. We consider quantile models that allow for dynamic effects from past quantiles and incorporate predictability from economic state variables. We choose the quantiles to represent different parts of the return distribution such as the left or right tails, center or 'shoulders'. Each quantile conveys valuable information. For example, the median can be used to capture location information, scale information can be obtained through the inter-quartile range and skewness and kurtosis through the difference between tail quantiles such as the 5% or 95% quantiles. Our approach thus generalizes existing measures that have focused on predictable patterns in moments such as the mean, variance, skew and kurtosis of returns. Given sufficiently many quantiles, we obtain a clear picture of how the return distribution depends on economic state variables.

Closely related to our paper is a literature that has focused on forecasting either the mean or the volatility of stock returns. Some papers have found evidence of predictability of mean returns using valuation ratios such as the earnings-price ratio or the dividend yield, interest rate measures

and a host of financial indicators such as corporate buybacks and payout ratios or macroeconomic variables such as inflation.¹ Findings of predictability of mean returns have been questioned, however, by Bossaerts and Hillion (1999) and Goyal and Welch (2003, 2007) who argue that the parameters of return prediction models are estimated with insufficient precision to make *ex-ante* return forecasts valuable.

While the volatility of stock returns is known to follow a pronounced counter-cyclical pattern (Schwert (1989)), there is relatively weak evidence that the level or volatility of macroeconomic state variables are helpful in predicting stock market volatility. Along with Schwert (1989), Engle, Ghysels and Sohn (2007) find some evidence that inflation volatility helps predict the volatility of stock returns. However, Engle, Ghysels and Sohn (2007) also find that the volatility of interest rate spreads and growth in industrial production, GDP or the monetary base fail to consistently predict future volatility, with evidence being particularly weak in the post-WWII sample. This is consistent with the findings in Paye (2006) and Ghysels, Santa-Clara and Valkanov (2006).

The difficulty experienced in establishing predictability of the conditional mean or variance through economic state variables does not imply that other parts of the return distribution cannot be predicted. To see this point, consider a simple prediction model relating monthly stock returns on the S&P500 index to the lagged default yield spread. Figure 1 compares the OLS estimate—which seeks to provide the best fit to expected returns—to estimates obtained using quantile regression. The horizontal axis lists quantiles running from 0.05 through 0.95, while coefficient estimates showing the effect of the state variable on the individual quantiles along with standard error bands are listed on the vertical axis. If the standard linear prediction model were true, the quantile estimates should, like the OLS coefficients, be constant across all quantiles and hence be flat lines. In fact, the quantile estimates follow a systematic pattern with large negative values in the left tail (for small quantiles) and large positive values in the right tail (for large quantiles). Moreover, whereas the OLS estimates fail to be significantly different from zero, the quantile estimates are mostly significant in the tails and ‘shoulders’ of the return distribution. The default spread thus appears to have little ability to predict the center (mean) of the return distribution but is capable of predicting tails of the return distribution. Clearly its failure to predict the mean return does *not*

¹A partial list of studies includes Ang and Bekaert (2007), Campbell (1987), Campbell and Shiller (1988), Campbell and Thompson (2007), Cochrane (2007), Fama and French (1988, 1989), Ferson (1990), Ferson and Harvey (1993), Lettau and Ludvigsson (2001) and Pesaran and Timmermann (1995).

imply that the default spread is not a valuable state variable for investors. This conclusion turns out to hold more generally: We find evidence that few of the state variables from the literature on predictability of stock returns can predict the center of the return distribution, but that many of these variables predict other parts of the return distribution.

The main contributions of our paper are as follows. First, we propose a quantile approach to capturing predictability in the distribution of stock returns. Our quantile prediction analysis offers many advantages over previous studies. By considering several quantiles, we gain flexibility to capture the ability of economic state variables to track predictability of different parts of the return distribution. Unlike estimates of higher order moments of returns, quantiles are robust to outliers which frequently affect stock returns (Harvey and Siddique (2000)) and can thus be estimated with greater precision than conventional moments of returns. Moreover, our approach is free of many of the parametric assumptions necessary when modeling the full return distribution. Finally, by considering sufficiently many quantiles, we obtain a relatively complete picture of time-variations in the return distribution which can be used for purposes of portfolio selection or asset pricing.

As our second contribution, we provide new and broader empirical evidence of predictability of US stock returns than previously available. We find that many of the state variables considered in the literature are useful in predicting either the left or right tails or ‘shoulders’ of the return distribution but not necessarily its center. For example, higher values of the smoothed earnings-price ratio or the term spread predominantly shift the upper quantiles of the return distribution to the right, thereby increasing the probability of surges in stock prices. Conversely, increased net equity expansion tends to precede large negative returns but has little ability to anticipate periods with large positive stock returns.

Our third contribution is to investigate the economic significance of predictability of return quantiles through an asset allocation exercise for an investor with power utility who combines stocks and T-bills. To this end we consider the out-of-sample asset allocation of an investor who uses our quantile forecasts to estimate the conditional return distribution. Gains from exploiting dynamic quantile forecasts in the asset allocation decisions appear to be sizeable in economic terms.

As our final contribution, we use our quantile models to predict events in the left and right tail of the distribution of stock returns. These predictions are compared to quantile forecasts implied by model-free options-based volatility estimates using the VIX contract traded on the Chicago Board Options Exchange (CBOE). This allows us to evaluate the information in the dynamic

quantile forecasts relative to market information embedded in options prices. If (after adjusting for a volatility risk premium) the dynamic quantile forecasts suggest a higher chance of large positive (negative) returns than indicated by the VIX estimates, we buy call (put) options. If the converse holds we sell options. Payoffs from these trades are compared with passive investments in the same options. Finally, to evaluate the economic significance of predictability in the tails, we use a second order stochastic dominance criterion which does not require specifying investors' preferences. Our findings provide evidence that a risk averse investor trading in options would find it beneficial to use the information embedded in the dynamic quantile forecasts.

The outline of the paper is as follows. Section 2 introduces the quantile approach to return predictability. Section 3 presents the data set and reports empirical results. Section 4 conducts an out-of-sample forecasting experiment and compares the performance of the proposed quantile models to alternatives from the existing literature. Section 5 evaluates the quantile forecasts in an asset allocation experiment, while Section 6 compares our quantile predictions to VIX-implied or Black-Scholes implied quantiles and investigates the economic value of the quantile forecasts through options trading. Finally, Section 7 concludes.

2 Modeling Quantiles of the Distribution of Stock Returns

Solving an expected utility maximizing investor's portfolio selection problem requires a model for the distribution of asset returns. Only in special cases such as under mean-variance preferences or normally distributed returns, are the first and second moments of the return distribution sufficient to solve this problem. In general, however, more detailed information on the return distribution is needed to solve for the optimal portfolio weights and characterize the risk of asset returns (Rothschild and Stiglitz (1970)).

To understand how different parts of the return distribution may depend on economic state variables, it is helpful to consider a range of quantiles located at separate points of the return distribution. Let $\alpha \in (0, 1)$ represent a particular quantile of interest. Varying α from values near zero (representing draws from the left tail of the return distribution) through middle values near one-half (representing the center) to values near one (representing the right tail) allows us to track variations in the complete return distribution. Moreover, by jointly considering a large number of quantiles, we can obtain a much richer picture of variations in the return distribution than is

available from the mean and variance. This can be used to indicate evidence of conditional skew or kurtosis and can also be used to uncover periods with the potential for unusually large negative or positive returns or to form confidence intervals for the return distribution.

An advantage of our approach is that it allows the effect of economic state variables to vary across quantiles whereas parametric models of the full return distribution tend to smooth the effect of state variables across different parts of the return distribution. When the effect of state variables on the return distribution is highly asymmetric, as we shall later see holds empirically in many cases, this is likely to lead to misspecified parametric models of the conditional return distribution.

We next describe our approach to modeling time variation in quantiles of the distribution of stock returns.

2.1 Quantile Models

A large literature in finance has explored whether the conditional mean or volatility of stock returns, r_{t+1} , vary through time as captured by models of the form

$$r_{t+1} = \mu_t + \sigma_t \varepsilon_{t+1}, \quad (1)$$

where μ_t and σ_t are the conditional mean and volatility, respectively, while ε_{t+1} is a return innovation with mean zero, variance one and a distribution function, F_ε , which is typically assumed to be time-invariant.

We are interested in analyzing whether state variables from the finance literature help predict parts of the return distribution beyond the mean and variance. To this end we model the conditional α -quantile of stock returns, denoted $q_\alpha(r_{t+1}|\mathcal{F}_t)$, where $\mathcal{F}_t$ contains information known at time t . For given values of the conditional mean and variance, the α -quantile of r_{t+1} implied by (1) is

$$q_\alpha(r_{t+1}|\mathcal{F}_t) = \mu_t + \sigma_t F_\varepsilon^{-1}(\alpha), \quad \alpha \in (0, 1). \quad (2)$$

For example, in the literature on predictability of mean returns it is common to assume that $\mu_t = \beta_0 + \beta_1 x_t$, where x_t represents predictor variables known at time t . In this case the quantile forecast becomes

$$q_\alpha(r_{t+1}|\mathcal{F}_t) = \beta_0 + \beta_1 x_t + \sigma_t F_\varepsilon^{-1}(\alpha). \quad (3)$$

If return innovations, ε , are symmetrically distributed, the median return forecast will be equal to

the mean return forecast:

$$E[r_{t+1}|\mathcal{F}_t] = q_{0.5}(r_{t+1}|\mathcal{F}_t) = \beta_0 + \beta_1 x_t. \quad (4)$$

This is the most common model from the literature on return predictability, see Goyal and Welch (2007). Such forecasts pertain to only one moment of the return distribution, namely its center.

There are good economic reasons, however, to explore if economic state variables can predict other parts of the return distribution. For example, evidence from different quantiles may help to interpret the economic source of return predictability and indicate whether it tracks time-varying risk, time-varying expected returns or perhaps even time-variations in the risk-return trade-off. Moreover, the type of return predictability may yield insights into how it can best be incorporated in investors' portfolio choice.

To explore predictability of the return distribution beyond the mean and variance, we consider a class of models that allows the individual return quantiles to depend on economic state variables, x_t :

$$q_\alpha(r_{t+1}|\mathcal{F}_t) = \beta_{0,\alpha} + \beta_{1,\alpha} x_t. \quad (5)$$

The local effect of x_t on the α -quantile is assumed to be linear. However, since we allow the slope coefficient ($\beta_{1,\alpha}$) to differ across quantiles, the model is very flexible.

This specification nests many existing models from the literature. First, the benchmark no-predictability model that assumes constant (time-invariant) return quantiles arises as a special case of (5) with $\beta_{1,\alpha} = 0$,

$$q_\alpha(r_{t+1}|\mathcal{F}_t) = \beta_{0,\alpha}. \quad (6)$$

Similarly, the standard prediction model where x_t simply shifts the conditional mean of the return distribution emerges when $\beta_{1,\alpha}$ does not vary across quantiles, i.e. $\beta_{1,\alpha} = \beta_1$ for all α :

$$q_\alpha(r_{t+1}|\mathcal{F}_t) = \beta_{0,\alpha} + \beta_1 x_t. \quad (7)$$

We next generalize (5) to allow for dynamic effects from past quantiles. To account for persistence in the distribution of stock returns, we follow Engle and Manganelli (2004) and include last period's conditional quantile and the absolute value of last period's return as predictor variables:

$$q_\alpha(r_{t+1}|\mathcal{F}_t) = \beta_{0,\alpha} + \beta_{1,\alpha} x_t + \beta_{2,\alpha} q_\alpha(r_t|\mathcal{F}_{t-1}) + \beta_{3,\alpha} |r_t|, \quad (8)$$

where $q_\alpha(r_t|\mathcal{F}_{t-1})$ is the lagged α -quantile and $|r_t|$ is the lagged absolute return. This specification is consistent with volatility clustering in stock returns.²

To gain intuition for the quantile models, note that if the effect of economic state variables on the return distribution arises through a volatility risk premium channel, we would expect to find the largest impact of such variables in the tails of the return distribution. To see this, suppose that return volatility varies in proportion with a state variable, x_t , and that it earns a risk premium, κ (see, e.g. Merton (1980)):

$$\begin{aligned} r_{t+1} &= \mu + \kappa\sigma_t + \sigma_t\varepsilon_{t+1}, & \varepsilon_{t+1} &\sim N(0, 1) \\ \sigma_t &= \varphi_0 + \varphi_1x_t, \end{aligned} \tag{9}$$

where φ_1 measures the volatility effect of x_t . This specification implies quantiles of the form

$$q_\alpha(r_{t+1}|\mathcal{F}_t) = \mu + \varphi_0(\kappa + q_{\alpha,N}) + (\kappa + q_{\alpha,N})\varphi_1x_t \equiv \beta_{0,\alpha} + \beta_{1,\alpha}x_t, \tag{10}$$

where the slope coefficient $\beta_{1,\alpha} = (\kappa + q_{\alpha,N})\varphi_1$ and $q_{\alpha,N}$ is the α -quantile of the normal distribution which takes on larger (absolute) values further out in the tails and shifts sign from negative to positive as α moves from values below the median to values above it. Economic theory suggests that $\kappa > 0$, so if we consider a variable with a positive correlation with volatility ($\varphi_1 > 0$), we should expect it to have negative slope coefficients in the quantile regression sufficiently far in the left tail (small α -values) and positive coefficients above the median. The reverse pattern should arise for variables correlating negatively with volatility ($\varphi_1 < 0$). We explore these effects in the empirical analysis in the next section.

2.2 Estimation

We estimate the parameters of the quantile prediction model as follows. Following Koenker and Bassett (1978), quantiles are estimated by replacing the conventional quadratic loss function underlying most empirical work on return predictability with the so-called ‘tick’ loss function

$$L_\alpha(e_{t+1}) = (\alpha - \mathbf{1}\{e_{t+1} < 0\})e_{t+1}, \tag{11}$$

²Foresi and Peracchi (1995) characterize the cumulative distribution function of stock returns as a function of a set of economic state variables. In effect they model the ‘dual’ of the quantile function and estimate conditional logit models over a grid of values for the cumulative distribution function of returns. There are many other differences, since we allow for autoregressive dynamics in the quantiles which are not considered by Foresi and Peracchi.

where $e_{t+1} = r_{t+1} - \hat{q}_{\alpha,t}$ is the forecast error, $\hat{q}_{\alpha,t} = q_{\alpha}(r_{t+1}|\mathcal{F}_t)$ is short-hand notation for the conditional quantile forecast computed at time t and $\mathbf{1}\{\cdot\}$ is the indicator function. Under this objective function, the optimal forecast is the conditional quantile. To see this, note that the first order condition associated with minimizing the expected value of (11) with respect to the forecast, $\hat{q}_{\alpha,t}$, is the α -quantile of the return distribution (see Koenker (2005))

$$\hat{q}_{\alpha,t} = F_t^{-1}(\alpha), \quad (12)$$

where F_t is the conditional distribution function of returns.

To obtain estimates of the parameters of the dynamic quantile specification in (8), we adopt the tick-exponential quasi maximum likelihood estimation approach proposed by Komunjer (2005) which extends the quantile regression method introduced by Koenker and Bassett (1978). Estimates of the parameters $\theta_{\alpha} = (\beta_{0,\alpha}, \beta_{1,\alpha}, \beta_{2,\alpha}, \beta_{3,\alpha})$ solve the objective

$$\hat{\theta}_{\alpha} = \arg \max_{\theta_{\alpha}} \left\{ T^{-1} \sum_{t=1}^T \ln \varphi_t^{\alpha}(r_t, q_{\alpha}(r_t|\mathcal{F}_{t-1}, \theta_{\alpha})) \right\}, \quad (13)$$

where φ_t^{α} is a probability density from the tick-exponential family:³

$$\varphi_t^{\alpha}(r_t, q_{\alpha}) = \exp\left(-\frac{1}{\alpha}(q_{\alpha} - r_t)\mathbf{1}\{r_t \leq q_{\alpha}\} + \frac{1}{1-\alpha}(q_{\alpha} - r_t)\mathbf{1}\{r_t > q_{\alpha}\}\right). \quad (14)$$

Komunjer (2005) establishes conditions under which the parameter estimates, $\hat{\theta}_{\alpha}$, are asymptotically normally distributed and provides methods for estimating their standard errors.⁴

3 Empirical Results

This section presents empirical results from applying the quantile models introduced in the previous section to explore in-sample predictability of US stock returns. In Section 4 we address out-of-sample predictability of the return distribution.

³In particular, we estimate the model using the minimax representation of the optimization problem. We use a special case of the tick exponential family which makes the objective function a constant times the tick loss function in (11).

⁴When estimating the dynamic quantile specification in (8) we restrict the parameter on the lagged quantile, $\beta_{2,\alpha}$, to lie between 0 and 1. To obtain an initial quantile, $q_{\alpha}(r_1|\mathcal{F}_0)$, we use the constant quantile as initial value and then estimate the model recursively.

3.1 Data

Our empirical analysis uses a data set comprising monthly stock returns along with a set of sixteen predictor variables previously analyzed in Goyal and Welch (2007).⁵ Stock returns are measured by the S&P500 index and include dividends. A short T-bill rate is subtracted from stock returns to obtain excess returns. The predictor variables we consider along with the data samples are listed in Table 1. The sample varies across variables with the longest data spanning the period 1871-2005, while the shortest sample covers the period 1937-2002. These long sample periods are important in order to get precise estimates of quantiles in the tails of the return distribution.

The predictor variables fall into four broad categories:

- Valuation ratios capturing some measure of ‘fundamental’ value to market value such as the
 - dividend-price ratio;
 - dividend yield;
 - earnings-price ratio;
 - 10-year earnings-price ratio;
 - book-to-market ratio;
- Bond yield measures capturing the level or slope of the term structure or measures of default risk, including the
 - three-month T-bill rate;
 - yield on long term government bonds;
 - term spread as measured by the difference between the yield on long-term government bonds and the three-month T-bill rate;
 - default yield spread as measured by the yield spread between BAA and AAA rated corporate bonds;
 - default return spread as measured by the difference between the yield on long-term corporate bonds and government bonds;

⁵We are grateful to Amit Goyal and Ivo Welch for providing this data.

- Estimates of equity risk such as the
 - cross-sectional equity premium, i.e. the relative valuations of high- and low-beta stocks;
 - long term return;
 - stock variance, i.e. a volatility estimate based on daily squared returns;
- Corporate finance variables, including the
 - dividend payout ratio measured by the log of the dividend-earnings ratio;
 - net equity expansion measured by the ratio of 12-month net issues by NYSE-listed stocks over their end-year market capitalization;

Finally, we also consider the inflation rate measured by the rate of change in the consumer price index. Additional details on data sources and the construction of these variables are provided by Goyal and Welch (2007).

3.2 Estimation Results

As a precursor to our quantile analysis, we first present full-sample estimates from OLS regressions of monthly stock returns on the individual predictor variables lagged one period. Table 2 shows that even at the 10% critical level only three of sixteen variables (inflation, the cross-sectional premium and net equity expansion) have significant predictive power over mean stock returns.

Since OLS estimates attempt to provide the best fit to the mean of the return distribution, we conclude from these results that predictability of the mean of US stock returns is rather weak. Only limited conclusions can be drawn from this evidence, however. In particular, we *cannot* conclude that the predictor variables fail to be useful for predicting other parts of the return distribution of interest to investors. For example, it could well be that a variable can predict events in the left tail (i.e. losses) although it fails to predict the center of the return distribution.

To explore this possibility, we next perform a series of quantile regressions for the univariate model in (5) which is the closest counterpart to the univariate linear regressions commonly used in the return predictability literature. Our analysis considers quantiles in the range $\alpha \in \{0.05, 0.10, 0.20, \dots, 0.90, 0.95\}$. Quantiles further out in the tails than 0.05 and 0.95 are not as precisely estimated and are hence not considered here.

Table 3 reports estimates of the slope coefficients ($\beta_{1,\alpha}$) for each of the predictor variables. Consistent with the weak evidence of predictability of mean returns only three variables generate significant slope coefficient at the median: Inflation and the T-bill rate are negatively related to median returns (consistent with findings for the mean reported by Fama and Schwert (1981) and Campbell (1987)) as is the payout ratio.

The standard linear return model (7) assumes that economic state variables have the same effect on the return distribution across all quantiles so $\beta_{1,\alpha} = \beta_1$ for all values of α . This is not the typical pattern found in Table 3. Many state variables work either in the tails but not in the center or they work in the left or right tail, but not in both. In fact, only two state variables, namely the stock variance and the default spread predict most (though not all) quantiles of the return distribution. Consistent with the risk story discussed earlier, the slope coefficients of both variables are generally greater in magnitude in the tails and switch signs from negative to positive. A rise in the default spread or stock variance is thus accompanied by an increased dispersion in future stock returns suggesting that these variables capture a predictable component in the riskiness of stock returns.

To gain intuition for this result, Figure 2 shows the quantiles of returns computed under three sets of values for the default spread: A middle scenario that sets this variable at its sample mean and scenarios where the default spread is set at its mean plus or minus two standard deviations. Increasing the default spread shifts the lower quantiles downwards and the upper quantiles upwards, reflecting a greater chance of large negative or large positive returns. Conversely, if the default yield is reduced, the lower (upper) quantiles of the return distribution are shifted upwards (downwards), thereby reducing the probability of large returns.

Variables such as the 10-year average earnings-price ratio, the payout ratio, the T-bill rate or net equity issues have asymmetric effects on the return distribution. For example, increased corporate (net) issues precede lower returns, moving the lower quantiles further to the left. Corporate issues do not appear to have a similar ability to predict surges in returns as reflected in the upper quantiles of the return distribution. This suggests that managers time their equity issues to precede periods with falling stock prices (Baker and Wurgler (2000)) although they cannot scale back issuing activity prior to periods with strongly increasing stock prices.

Higher T-bill rates seem mainly to reduce the central and upper quantiles of the return distribution without having a similar effect on the lower quantiles. Low T-bill rates are thus associated

with strong market performance, while conversely high T-bill rates do not augur bear markets.

To address if a particular state variable helps forecast some part of the return distribution, the last column of Table 3 reports Bonferroni p -values. These provide a summary measure of whether a given predictor variable is significant across any of the quantiles considered jointly and are robust to arbitrary dependencies across individual quantiles. By this criterion, close to half of the state variables are significant at the 5% critical level. This evidence stands in marked contrast to the earlier findings in Table 2 revealing weak (in-sample) predictability of the mean of stock returns.

We conclude that although most valuation ratios (e.g. the dividend yield or the earnings-price ratio) fail to predict any part of the return distribution, many of the predictor variables proposed in the finance literature, including the T-bill rate, inflation, the default yield, stock variance, payout ratio and net equity issues contain valuable information for predicting parts of the return distribution.

3.3 Quantiles and Higher Moments Of the Return Distribution

To assist with the economic interpretation of our results we next study how the conditional quantiles evolve over time. This achieves two objectives. First, it allows us to see how extensive the variation in the predicted quantiles is over time and whether return predictability varies across quantiles. Second, it allows us to link movements in the quantiles to specific historical events, which provides another way of assessing the information embodied in the quantile forecasts.

Figure 3 plots the 5%, 10%, 50%, 90% and 95% quantiles over the period 1970-2005 based on the dynamic quantile specification (8) that uses the default yield spread as a state variable. Horizontal lines show the corresponding quantiles based on the model that assumes constant quantiles.⁶

There is considerable variation over time in the conditional quantiles. Moreover, as witnessed by the frequent widening in both the lower and upper quantiles, this variation is highly persistent and much stronger in the tails than at the median. Some patterns in return predictability are clearly volatility driven. This includes the period following the oil price shocks of 1974/75 and a six-month period after the stock market crash of October 1987. Both episodes were associated with highly uncertain market conditions.

⁶Despite their proximity there are very few crossings between the 90% and 95% quantile estimates or between the 5% and 10% quantile estimates. This is to be expected if our quantile model is correctly specified since $\hat{q}_{\alpha_1} < \hat{q}_{\alpha_2}$ for $\alpha_1 < \alpha_2$, even though we do not impose this restriction in our estimation.

At other times the lower tail quantiles decline significantly more than the upper quantiles rise, indicating substantial downside risk. This happens during the period from November 1979 to May 1980 following the change in the Federal Reserve's monetary policy and again in mid-1994 and in 1996. The reverse scenario—a significant increase in the upper quantiles without a corresponding fall in the lower quantiles—is seen in 1983 and 1986. Both scenarios indicate important asymmetries in the return distribution. Clearly there is much more to the variation in the quantiles than can be accounted for by time-varying volatility alone.

Figure 3 reveals very different persistence of the quantiles in the tails and center of the return distribution. This is due, in part, to the different slope coefficients of the economic state variables in the tails (large values) versus the center (small values). However, it also reflects different patterns in the slope coefficients of the lagged quantile and lagged absolute returns. To see this, Figure 4 plots the coefficient estimates of the lagged quantile ($\beta_{2,\alpha}$) and the lagged absolute return ($\beta_{3,\alpha}$) for the dynamic quantile model. The left window reveals a high persistence for both lower ($\alpha \leq 0.3$) and upper ($\alpha \geq 0.6$) quantiles but very little persistence in the center. Similarly, lagged absolute returns have a significant negative effect on the lower quantiles ($\alpha \leq 0.3$) and a significant positive effect on the upper quantiles ($\alpha \geq 0.6$) but little effect in the center. Since the absolute values of returns are quite persistent, again this is consistent with the higher persistence observed in the tails than in the center of the return distribution.

Quantiles capture different parts of the return distribution and can be used as the basis for shape measures such as skew and kurtosis which have been shown to have important implications for investors' portfolio allocation (Harvey, Liechty and Liechty (2004), Guidolin and Timmermann (2006)). Higher moments also appear to have implications for the cross-section of stock returns in the sense that exposure to negative skew or downside risk of the market portfolio earns a risk premium (Harvey and Siddique (2000), Dittmar (2002) and Ang, Chen and Xing (2006)).

Measures of the shape of the stock return distribution such as the skew and kurtosis are typically estimated directly from sample observations on returns raised to the third and fourth power, respectively. This has the effect of increasing the sensitivity of the estimates to outliers and hence increases estimation error. This is even more of a concern when the moments are estimated conditionally in order to get a sense of time-variation in higher order moments.

To deal with this problem, robust quantile-based measures of skewness and kurtosis have been proposed. Extending the measure of skewness proposed by Bowley (1920) to the conditional case,

we get

$$\widehat{SK}_t = \frac{\hat{q}_{0.75,t} + \hat{q}_{0.25,t} - 2\hat{q}_{0.5,t}}{\hat{q}_{0.75,t} - \hat{q}_{0.25,t}}. \quad (15)$$

Differences in the distance between the first quartile and the median and the distance between the third quartile and the median are used here to capture skews in the return distribution. Similarly, building on the kurtosis measure proposed by Crow and Siddiqui (1967), centered so as to be zero under the Gaussian distribution, we use the following conditional kurtosis measure:

$$\widehat{KR}_t = \frac{\hat{q}_{1-\alpha,t} - \hat{q}_{\alpha,t}}{\hat{q}_{1-\beta,t} - \hat{q}_{\beta,t}} - 2.91. \quad (16)$$

In our calculations we follow Kim and White (2003) and set α and β to 0.025 and 0.25, respectively.

Figure 5 plots the time series of conditional skewness based on the dynamic quantile model that includes the default yield spread as a predictor variable. The return distribution is negatively skewed most of the time although there were periods around the mid-eighties and during the mid-to-late nineties where the return distribution became right-skewed in anticipation of an ensuing rise in market prices. The strongest negative skew appeared after the oil shocks in the mid-seventies, in the early eighties (during the change in monetary policy), after 1987 and during the bear market, 2000-2003.

Conversely, the conditional excess kurtosis of the return distribution, plotted in Figure 6, is largely positive with peaks around the same periods where the return distribution has a negative skew, signalling greater risks during those points in time.

We conclude from these plots of the skew and kurtosis that our time-varying quantile estimates are highly informative for capturing changes in the conditional higher order moments of the stock return distribution. Unlike conventional measures, our estimates are not greatly affected by outliers in returns.

4 Does Any Variable Predict Return Quantiles Out-of-sample?

Ex-ante or out-of-sample predictability of stock returns remains an extensively debated question. While many studies have documented in-sample predictability of mean returns, Goyal and Welch (2003, 2007) find little evidence to suggest that expected returns can be predicted out-of-sample by any of the variables considered here, a conclusion supported by the evidence in Bossaerts and Hillion (1999) and Lettau and van Niekerk (2007). Still, studies such as Pesaran and Timmermann

(1995), Ang and Bekaert (2007) and Campbell and Thompson (2007) find some evidence of out-of-sample predictability of the conditional mean. Here we address a new question, namely the out-of-sample predictability of the full return distribution. We first set out to do this using statistical criteria. Sections 5 and 6 provide more direct economic measures of out-of-sample forecasting performance.

4.1 Forecasting Performance

To evaluate the forecasting performance of our quantile models out-of-sample, we estimate the parameters of the quantile prediction models using data from the start of the sample up to 1969:12. One-step-ahead forecasts are then generated for returns in 1970:01. The following period we update our estimates by adding data from 1970:01 and use the updated model to produce quantile forecasts for 1970:02. This recursive forecasting procedure is repeated up to the end of the sample generating a set of 432 out-of-sample forecasts for the period 1970:01-2005:12. This can be considered a challenging sample period as it includes the oil shocks and stagflation period of the 1970s, the shift in monetary policy from 1979-82, the stock market bubble of the 1990s and the ensuing downturn.

We present results for four quantile forecasting models, namely (i) the dynamic quantile specification ((8)) based on each of the individual predictor variables; (ii) an equal-weighted combination of the forecasts from each of the univariate quantile models computed as $\bar{q}_{\alpha,t} = (1/16) \sum_{i=1}^{16} \hat{q}_{\alpha,t}^i$, where $\hat{q}_{\alpha,t}^i$ is the conditional α -quantile associated with model i . This provides a way to incorporate multivariate information from the individual quantile forecasts without having to estimate additional parameters. Such simple averages have proved difficult to outperform in a variety of settings in economics and finance (Timmermann (2006));⁷ (iii) a GARCH(1,1) specification which captures predictability in the volatility of stock returns; (iv) a constant or ‘prevailing’ quantile (PQ) model with no predictor variables (6). This is the obvious ‘no predictability’ counterpart to the prevailing mean model used by Goyal and Welch (2007).

As a first measure of model ‘fit’, Table 4 reports out-of-sample coverage ratios, i.e. the percentage of times that actual returns fall below the predicted α -quantile for $\alpha = \{0.05, 0.1, 0.5, 0.9, 0.95\}$. For most of the quantile models the coverage ratios are close to their correct values, i.e. roughly 5% of stock returns fall below the 5% quantile forecasts, roughly 10% of returns fall below the

⁷Maheu and McCurdy (2007) also find that submodel averaging leads to improved models of the unconditional distribution of stock returns.

10% quantile forecasts etc. This also holds on average as witnessed by the performance of the equal-weighted quantile combination and holds as well for the GARCH and PQ models. On this criterion at least, none of the quantile prediction models appears to be obviously misspecified.

To assess whether any of the dynamic quantile models performs better than the constant or ‘prevailing’ quantile model, Table 5 reports out-of-sample average loss for the models under consideration. This comparison uses the tick objective function (11) and thus provides a statistical measure of predictive accuracy based on the models’ ability to predict if returns fall above or below a particular point. Studies such as Leitch and Tanner (1991) have found that this type of loss function is more closely related to the possibility of making economic profits from return forecasts than conventional measures such as mean squared error.

Univariate quantile models struggle in the left tails ($\alpha = 0.05$ and $\alpha = 0.10$) where only three and five out of sixteen models improve upon the results produced by the simple prevailing quantile model which assumes a constant return distribution. Even worse performance is observed in the center of the return distribution where only three of sixteen univariate quantile models come out on top of the prevailing quantile model. This can be explained by the greater parameter estimation errors associated with the dynamic quantile models compared with the constant quantile model.

Very different results emerge in the right tail of the return distribution. For $\alpha = 0.9$ and $\alpha = 0.95$, thirteen out of sixteen quantile models produce lower out-of-sample average loss than the prevailing quantile method.

Averaging quantile forecasts across different predictor variables seems to add value as the simple equal-weighted quantile forecasts work very well. With only one exception, the equal-weighted quantile forecasts always generate lower out-of-sample loss than both the prevailing quantile and GARCH(1,1) quantile forecasts. Moreover, the simple equal-weighted quantile forecast improves upon the vast majority of the individual univariate quantile forecasts, most likely due to the benefits of including information from multiple predictor variables.

4.2 Significance of Time-Varying Quantiles

To explore if any of the dynamic quantile prediction models add significant information beyond the ‘prevailing quantile’ forecasts, we consider the weights on the univariate dynamic quantiles versus those on the prevailing quantile in a combined forecast. If the weights on the time-varying quantile forecasts are non-zero, we can conclude that they provide valuable information. The closer these

weights are to one, the stronger is the evidence that the time-varying quantile forecasts add value beyond the prevailing quantiles.

Let $\hat{q}_{\alpha,t}^{DQ}$ be the quantile forecast produced by the dynamic model (8), while $\hat{q}_{\alpha,t}^{PQ}$ is the corresponding prevailing quantile forecast based on (6). We are interested in testing whether information embedded in $\hat{q}_{\alpha,t}^{DQ}$ helps improve on the forecasting performance of the prevailing quantile model. To this end we consider the combined quantile forecast

$$\hat{q}_{\alpha,t}^c = \lambda_{\alpha}^0 + \lambda_{\alpha}^{DQ} \hat{q}_{\alpha,t}^{DQ} + \lambda_{\alpha}^{PQ} \hat{q}_{\alpha,t}^{PQ} \quad (17)$$

and test whether $\tilde{\lambda}_{\alpha}^{DQ} = 0$, where

$$(\tilde{\lambda}_{\alpha}^0, \tilde{\lambda}_{\alpha}^{DQ}, \tilde{\lambda}_{\alpha}^{PQ}) = \arg \min_{\lambda_{\alpha}^0, \lambda_{\alpha}^{DQ}, \lambda_{\alpha}^{PQ}} E_t[L_{\alpha}(r_{t+1} - \lambda_{\alpha}^0 - \lambda_{\alpha}^{DQ} \hat{q}_{\alpha,t}^{DQ} - \lambda_{\alpha}^{PQ} \hat{q}_{\alpha,t}^{PQ})], \quad (18)$$

where $L_{\alpha}(\cdot)$ is the tick loss function in (11).

The first order condition associated with this equation implies that the vector of optimal combination weights $\tilde{\lambda}_{\alpha} = (\tilde{\lambda}_{\alpha}^0, \tilde{\lambda}_{\alpha}^{DQ}, \tilde{\lambda}_{\alpha}^{PQ})'$ satisfies

$$E_t[\alpha - \mathbf{1}\{r_{t+1} - \tilde{\lambda}_{\alpha}^0 - \tilde{\lambda}_{\alpha}^{DQ} \hat{q}_{\alpha,t}^{DQ} - \tilde{\lambda}_{\alpha}^{PQ} \hat{q}_{\alpha,t}^{PQ}\} < 0] = 0. \quad (19)$$

From these moment conditions, estimates of $\lambda_{\alpha} = (\lambda_{\alpha}^0, \lambda_{\alpha}^{DQ}, \lambda_{\alpha}^{PQ})'$ can be obtained via the generalized method of moments (GMM) using a vector of instruments z_t and sample moments

$$\frac{1}{T} \sum_{t=1}^T g(\lambda_{\alpha}; r_{t+1}, z_t) = \frac{1}{T} \sum_{t=1}^T [\alpha - \mathbf{1}\{r_{t+1} - \lambda'_{\alpha} \hat{q}_{\alpha,t} < 0\}] z_t, \quad (20)$$

where $\hat{q}_{\alpha,t} = (1, \hat{q}_{\alpha,t}^{DQ}, \hat{q}_{\alpha,t}^{PQ})'$. We use a constant, the lagged covariate, the lagged return and lagged quantile forecasts as instruments except for the equal-weighted forecast combination where the lagged covariate is dropped.⁸

⁸The asymptotic distribution of the GMM estimates of λ_{α} requires that the moment conditions are once differentiable. Since the indicator function in the moment condition (19) poses a problem, we follow Giacomini and Komunjer (2005) by replacing $g(\lambda_{\alpha}; r_{t+1}, z_t)$ with the following smooth approximation:

$$g(\lambda_{\alpha}; r_{t+1}, z_t, \tau) = [\alpha - (1 - \exp(-(r_{t+1} - \lambda'_{\alpha} \hat{q}_{\alpha,t})/\tau)) \mathbf{1}\{r_{t+1} - \lambda'_{\alpha} \hat{q}_{\alpha,t} < 0\}] z_t.$$

Here τ is a smoothing parameter which is set equal to 0.005. GMM estimation of the combination weights, λ_{α} , is carried out recursively using a heteroskedasticity and autocorrelation consistent weighting matrix. Recursive GMM estimation of optimal forecast combination weights requires choices of instruments, initial combination weights and an initial weighting matrix. The initial weighting matrix is always set to the identity matrix whereas we conduct a

Table 6 reports empirical estimates of the combination weights when we apply GMM estimation to our data. In the left tail ($\alpha = 0.05$ and $\alpha = 0.10$) and the center ($\alpha = 0.50$) of the return distribution there are few cases with significant weights on the time-varying quantile predictions. Conversely, there are many instances where the weight on the prevailing quantile forecasts are significant at the 10% level (e.g., in 11 of 16 cases for $\alpha = 0.05$ and $\alpha = 0.10$).

Very different conclusions emerge for the right tail of the return distribution ($\alpha = 0.90$ and $\alpha = 0.95$) where virtually all of the dynamic quantile forecasts generate significant weights. Moreover, these weights are frequently quite large and always positive. These findings strongly suggest that it is possible to use economic state variables to produce better *ex-ante* forecasts of upper return quantiles than those associated with the prevailing quantile model which assumes no predictability.

The final row in Table 6 compares the out-of-sample performance of the equal-weighted quantile forecasts to that produced by forecasts based on the prevailing quantile model. As revealed by their large values close to one, the equal-weighted quantile forecasts dominate prevailing quantile forecasts in the upper parts of the return distribution, i.e. for $\alpha = 0.90$ and $\alpha = 0.95$. There is also some evidence that the equal-weighted quantile forecast dominates the prevailing quantile for $\alpha = 0.05$.

We conclude from this analysis that, using statistical measures of forecast accuracy, there is substantial evidence that including information in economic state variables through dynamic quantile models helps predict time-variations in the distribution of stock returns in a way that the prevailing quantile model does not facilitate.

5 Economic Significance

To evaluate the economic significance of the information embedded in our quantile predictions of stock returns, we next consider their use in the out-of-sample asset allocation decisions of a risk averse investor with power utility.

global search for the best initial combination weights. We first generate 5000 random combination weights from a uniform distribution on $[-2,2]$ and choose those 500 initial values with the smallest loss. We then estimate the optimal forecast combination weights via GMM for each of these 500 initial values and report the combination weights that generate the smallest value of the minimized objective function.

5.1 Portfolio Selection

Consider an investor who allocates $w_t W_t$ of total wealth to stocks and the remainder, $(1 - w_t)W_t$ to a risk-free asset, where W_t is the initial wealth in period t . Without loss of generality we set $W_t = 1$ so the wealth at time $t + 1$ is given by

$$\begin{aligned} W_{t+1} &= 1 + r_t^f + w_t(r_{t+1}^s - r_t^f) \\ &\equiv 1 + r_t^f + w_t \rho_{t+1}, \end{aligned}$$

where ρ_{t+1} is the return on the stock market index in excess of the risk-free rate, r_t^f . Following standard practice, we assume the investor is small and has no market impact. Moreover, we assume that the investor has power utility over terminal wealth,

$$U(W_{t+1}) = \frac{W_{t+1}^{1-\gamma}}{1-\gamma}, \quad (21)$$

where γ is the investor's coefficient of relative risk aversion. Portfolio weights for period t can be obtained as the solution to the following optimization problem:

$$w_t^* = \arg \max_{w_t} E_t[\beta U(W_{t+1})], \quad (22)$$

where β is a subjective discount factor and $E_t[\cdot]$ denotes the conditional expectation based on the investor's information set in period t . In a given period, we assume that the investor solves equation (22), holds the optimal portfolio for one period and then reoptimizes the portfolio weights the following period based on new information. We set the investment horizon to one period and ignore any intertemporal hedging component in the investor's portfolio choice. The portfolio optimization problem in (22) can be written as

$$w_t^* = \arg \max_{w_t} \int \frac{\beta}{1-\gamma} (1 + r_t^f + w_t \rho_{t+1})^{1-\gamma} f(\rho_{t+1} | \mathcal{F}_t) d\rho_{t+1}, \quad (23)$$

where $f(\rho_{t+1} | \mathcal{F}_t)$ is the conditional probability distribution of future excess returns based on the investor's information set in period t . To solve for the optimal weights, w_t^* , the investor thus needs an estimate of the conditional distribution of future (excess) stock returns.

We obtain this by using our quantile forecasts to approximate $f(\rho_{t+1} | \mathcal{F}_t)$ by assuming that stock returns in period $t + 1$ are piecewise uniformly distributed between the quantile forecasts formed in period t with exponentially decaying tails. Specifically, let $\tilde{q}_{\alpha,t}$ denote the forecast of the

α -quantile of the excess return distribution in period $t + 1$ based on the information set in period t . We assume that the distribution can be approximated by

$$f(\rho_{t+1}|\mathcal{F}_t) = \begin{cases} \frac{1}{\sqrt{2\pi}\tilde{\sigma}_t} \exp\left(\frac{-(\rho_{t+1}-\tilde{\mu}_t)^2}{2\tilde{\sigma}_t^2}\right), & \text{if } \rho_{t+1} \leq \tilde{q}_{0.05,t} \\ \frac{0.05}{\tilde{q}_{0.10,t}-\tilde{q}_{0.05,t}}, & \tilde{q}_{0.05,t} \leq \rho_{t+1} \leq \tilde{q}_{0.10,t} \\ \frac{0.1}{\tilde{q}_{\alpha+0.10,t}-\tilde{q}_{\alpha,t}}, & \tilde{q}_{\alpha,t} \leq \rho_{t+1} \leq \tilde{q}_{\alpha+0.10,t} \\ & (\alpha \in [0.10, 0.80]) \\ \frac{0.05}{\tilde{q}_{0.95,t}-\tilde{q}_{0.90,t}}, & \tilde{q}_{0.90,t} \leq \rho_{t+1} \leq \tilde{q}_{0.95,t} \\ \frac{1}{\sqrt{2\pi}\tilde{\sigma}_t} \exp\left(\frac{-(\rho_{t+1}-\tilde{\mu}_t)^2}{2\tilde{\sigma}_t^2}\right), & \text{if } \rho_{t+1} > \tilde{q}_{0.95,t} \end{cases} \quad (24)$$

where $\tilde{\mu}_t$ and $\tilde{\sigma}_t$ are estimates of the center and dispersion of the return distribution which ensure that the return distribution is continuous at the 5% and 95% quantiles.⁹

Using this expression for the conditional return distribution, the portfolio optimization problem in (23) can be written as:

$$\begin{aligned} w_t^* &= \arg \max_{w_t} \\ &\int_{-\infty}^{\tilde{q}_{0.05,t}} \frac{\beta}{1-\gamma} (1+r_t^f + w_t \rho_{t+1})^{1-\gamma} \frac{1}{\sqrt{2\pi}\tilde{\sigma}_t} \exp(-(\rho_{t+1}-\tilde{\mu}_t)^2/2\tilde{\sigma}_t^2) d\rho_{t+1} \\ &+ \int_{\tilde{q}_{0.05,t}}^{\tilde{q}_{0.10,t}} \frac{\beta}{1-\gamma} (1+r_t^f + w_t \rho_{t+1})^{1-\gamma} \frac{0.05}{(\tilde{q}_{0.10,t}-\tilde{q}_{0.05,t})} d\rho_{t+1} \\ &+ \sum_{\alpha=0.1}^{0.8} \int_{\tilde{q}_{\alpha,t}}^{\tilde{q}_{\alpha+0.10,t}} \frac{\beta}{1-\gamma} (1+r_t^f + w_t \rho_{t+1})^{1-\gamma} \frac{0.1}{(\tilde{q}_{\alpha+0.10,t}-\tilde{q}_{\alpha,t})} d\rho_{t+1} \\ &+ \int_{\tilde{q}_{0.90,t}}^{\tilde{q}_{0.95,t}} \frac{\beta}{1-\gamma} (1+r_t^f + w_t \rho_{t+1})^{1-\gamma} \frac{0.05}{(\tilde{q}_{0.95,t}-\tilde{q}_{0.90,t})} d\rho_{t+1} \\ &+ \int_{\tilde{q}_{0.95,t}}^{+\infty} \frac{\beta}{1-\gamma} (1+r_t^f + w_t \rho_{t+1})^{1-\gamma} \frac{1}{\sqrt{2\pi}\tilde{\sigma}_t} \exp(-(\rho_{t+1}-\tilde{\mu}_t)^2/2\tilde{\sigma}_t^2) d\rho_{t+1}. \end{aligned} \quad (25)$$

All the middle terms in the portfolio optimization problem (25) can be integrated analytically whereas the first and last terms need to be solved numerically for a given w_t . Incorporating the

⁹Instead of assuming uniform distributions between the individual quantiles, we also considered Gaussian kernels for the probability distribution between the individual quantiles. This approach yielded very similar results.

analytical solutions to the integrals, the portfolio optimization problem simplifies to

$$\begin{aligned}
w_t^* = \arg \max_{w_t} & \int_{-\infty}^{\tilde{q}_{0.05,t}} \frac{\beta}{1-\gamma} (1+r_t^f + w_t \rho_{t+1})^{1-\gamma} \frac{1}{\sqrt{2\pi}\tilde{\sigma}_t} \exp(-(\rho_{t+1} - \tilde{\mu}_t)^2/2\tilde{\sigma}_t^2) d\rho_{t+1} \\
& + \frac{\beta}{(1-\gamma)(2-\gamma)w_t} \left[\frac{0.05}{(\tilde{q}_{0.10,t} - \tilde{q}_{0.05,t})} [(1+r_t^f + w_t \tilde{q}_{0.10,t})^{2-\gamma} - (1+r_t^f + w_t \tilde{q}_{0.05,t})^{2-\gamma}] \right. \\
& + \sum_{\alpha=0.1}^{0.8} \frac{0.1}{(\tilde{q}_{\alpha+0.10,t} - \tilde{q}_{\alpha,t})} [(1+r_t^f + w_t \tilde{q}_{\alpha+0.10,t})^{2-\gamma} - (1+r_t^f + w_t \tilde{q}_{\alpha,t})^{2-\gamma}] \\
& + \left. \frac{0.05}{(\tilde{q}_{0.95,t} - \tilde{q}_{0.90,t})} [(1+r_t^f + w_t \tilde{q}_{0.95,t})^{2-\gamma} - (1+r_t^f + w_t \tilde{q}_{0.90,t})^{2-\gamma}] \right] \\
& + \int_{\tilde{q}_{0.95,t}}^{+\infty} \frac{\beta}{1-\gamma} (1+r_t^f + w_t \rho_{t+1})^{1-\gamma} \frac{1}{\sqrt{2\pi}\tilde{\sigma}_t} \exp(-(\rho_{t+1} - \tilde{\mu}_t)^2/2\tilde{\sigma}_t^2) d\rho_{t+1}. \tag{26}
\end{aligned}$$

where $\gamma \neq 1, 2$ and $w_t \neq 0$. The analytical solution to the middle integrals takes the following form for a log-utility investor:

$$\begin{aligned}
& \int \beta \log(1+r_t^f + w_t \rho_{t+1}) f(\rho_{t+1}|\mathcal{F}_t) d\rho_{t+1} \\
& = \int \beta \log(1+r_t^f + w_t \rho_{t+1}) \frac{k_\alpha}{\tilde{q}_{\alpha+k_\alpha,t} - \tilde{q}_{\alpha,t}} d\rho_{t+1} \\
& = \frac{\beta k_\alpha}{\tilde{q}_{\alpha+k_\alpha,t} - \tilde{q}_{\alpha,t}} \left[\left(\frac{1+r_t^f}{w_t} + \rho_{t+1} \right) \log \left(\frac{1+r_t^f}{w_t} + \rho_{t+1} \right) + (\log(w_t) - 1) \rho_{t+1} \right],
\end{aligned}$$

where $k_\alpha = 0.05$ for $\alpha = \{0.05, 0.90\}$ and $k_\alpha = 0.10$ for $\alpha = \{0.10, 0.20, \dots, 0.80\}$.¹⁰

Each period the investor chooses the optimal portfolio weight, w_t^* , by solving (26) using quantile forecasts of the return distribution. To rule out short sales, we restrict the optimal portfolio weights to lie between zero and one. Moreover, we calculate the outer integrals in (26) numerically.

The resulting portfolio weights, w_t^* , give rise to a realized utility next period of $U(W_{t+1}^*) = (1+r_t^f + w_t^* \rho_{t+1})^{1-\gamma}/(1-\gamma)$. We assess the economic value of the quantile forecasts through the associated certainty equivalence return (CER):

$$CER = \left((1-\gamma)T^{-1} \sum_{t=1}^T U(W_t^*) \right)^{1/(1-\gamma)} - 1, \tag{27}$$

where $1/T \sum_{t=1}^T U(W_t^*)$ is the mean realized utility and T is the total number of observations in the out-of-sample period.

¹⁰For $\gamma = 2$ and $w_t = 0$, the closed form solutions to the integrals are obtained by taking limits. The solutions are available from the authors upon request.

5.2 Empirical Results

Table 7 presents empirical results based on our out-of-sample quantile forecasts over the period 1970-2005. We consider three levels of risk aversion, namely $\gamma = 1$ (log utility or low risk aversion), $\gamma = 2.5$ (corresponding to medium risk aversion) and $\gamma = 5$ (high risk aversion). First consider the results under logarithmic utility. For this case 9 of 16 univariate quantile prediction models yield higher CER values than the prevailing quantile model (PQ). Gains range from small improvements up to 3% per annum in the case of the term spread. The highest CER values are associated with the dynamic quantile forecasts that use inflation, the T-bill rate, term spread, long-term yield or long-term return as predictor variables. Investments based on forecasts from these models all improve on the CER of the PQ model by more than 1% per annum. Moreover, whereas the GARCH model is dominated by the PQ model, the equal-weighted quantile forecasts perform very well, producing a gain in the CER-value of nearly 2% over the constant distribution model.

Turning to the medium risk aversion case ($\gamma = 2.5$), the dynamic quantile forecasts based on the T-bill rate, long-term yield, term spread, cross-sectional premium, long-term return and inflation continue to produce higher CER-values than the PQ model. Moreover, the CER-value associated with the equal-weighted forecast combination exceeds that of the PQ model by more than 70 basis points per annum.

Finally, when γ is raised to 5, the investor becomes more risk averse and hence is less inclined to exploit time-variations in the return distribution. This has the effect of dampening gains from information embedded in the dynamic quantile forecasts. Still, many of the univariate models continue to outperform the PQ model as does the equal-weighted average which produces a gain in the CER of nearly 40 basis points per annum relative to the benchmark.¹¹

We conclude that the evidence of predictable time-variations in the distribution of stock returns is sufficiently strong to be of economic value to a risk-averse investor. Moreover, when quantile forecasts from the univariate models are combined into a simple equal-weighted average, the resulting forecast produces higher certainty equivalent returns across different levels of risk aversion.

¹¹Over a three-year out-of-sample period from 1989 to 1992, Foresi and Peracchi (1995) also find some evidence that their forecasts of the cumulative distribution of stock returns can be used to improve investment returns.

6 Option Trading Strategies

Our results thus far indicate that conditional quantile forecasts are of particular value in the tails of the return distribution and less so in its center. This raises the question of how investors can best exploit such information. It is natural to consider options whose strikes are selected to match predictability in the tails. We therefore next review a range of option trading strategies based on comparisons between dynamic quantile forecasts and quantile forecasts implied either by the Chicago Board of Exchange (CBOE) Volatility Index (VIX), which we refer to as VIX quantiles, or by the Black-Scholes implied volatility calculated using at-the-money S&P 500 options, which we refer to as IV quantiles.

6.1 Option-Implied Quantiles

Since its introduction by the CBOE in 1993, the VIX has been considered a leading measure of the market's near term volatility. It is a measure of market expectations of future volatility of the S&P 500 index implied by the options trading on this index. The VIX derives the expected volatility by averaging the weighted prices of a range of out-of-the-money puts and calls. For our purpose, the most important feature of the VIX is that it is model independent. This has several advantages. First, as it uses a weighted average of several option prices, it is a more robust measure than implied volatility from the Black-Scholes option pricing model.¹² Second, the VIX provides a measure of volatility close to those used by financial theorists and market practitioners, in part because it is valid under a broad set of assumptions on the dynamics governing stock returns.

Our data consists of daily observations on S&P 500 index options (SPX) between 1990 and 2005. The data contains contract information on all available S&P 500 options such as the type of the option (put or call), expiration date and strike price as well as open interest and volume. To be consistent with our return forecasts, we focus only on observations on the last trading day of each month. An option with an end-of-month expiration day would be ideal for our empirical study since this is the period used by our quantile models. However, the S&P 500 index options expire on Saturdays following the third Friday of the expiration month. The nearest term option therefore, on average, has a time to maturity of 17 days from the end of the month while the second nearest term option has a time to maturity of 45 days.

¹²See Andersen and Bondarenko (2007) for a discussion of some limitations of the VIX and suggested refinements.

To obtain VIX-implied quantile forecasts of returns in a given month, we assume that the excess return distribution is centered on the prevailing mean with a standard deviation implied by the VIX on the last trading day of the previous month. Formally, let $\bar{\mu}_t$ denote the prevailing mean estimate of monthly excess returns at time t , and let $\hat{\sigma}_{VIX,t} = VIX_t/\sqrt{12}$ denote the volatility implied by the VIX, where VIX_t is the closing value of the VIX on the last trading day of month t . Assuming that continuously compounded returns are normally distributed, the forecast of the α -quantile of returns in month $t + 1$ given information at time t , $\hat{q}_{\alpha,t}^{VIX}$, is calculated as follows:

$$\hat{q}_{\alpha,t}^{VIX} = \bar{\mu}_t + q_{\alpha,N} \hat{\sigma}_{VIX,t}, \quad (28)$$

where $q_{\alpha,N}$ is the α -quantile of the normal distribution. For comparison, a forecast of the Black-Scholes implied α -quantile of returns in month $t + 1$ given information at time t , $\hat{q}_{\alpha,t}^{IV}$, is calculated as follows:

$$\hat{q}_{\alpha,t}^{IV} = \bar{\mu}_t + q_{\alpha,N} \hat{\sigma}_{IV,t}, \quad (29)$$

where $\hat{\sigma}_{IV,t}$ is the monthly implied volatility computed from the S&P 500 index options.¹³

We compare these option-implied quantile forecasts to the time-varying quantile predictions obtained from the equal-weighted quantile combination based on information available at the end of month t .

Because the VIX or implied volatility seek to measure the expected integrated variance under the risk-neutral measure, they are not directly comparable to our quantile forecasts which are computed under the actual, or objective, probability measure. In other words, the VIX or implied volatility cannot be interpreted as pure volatility forecasts but are a combination of a volatility forecast and a risk premium for the uncertainty surrounding future market volatility. While our quantile forecasts and the option-implied quantiles therefore cannot be directly compared, they can be expected to respond to the same sort of information about future volatility. Rather than modeling how the volatility risk premium evolves over time, we estimate its average value and consider whether the dynamic quantile forecasts, $\hat{q}_{\alpha,t}^{DQ}$, differ from the option-implied quantiles by more than their average historical difference, whose sample estimate we denote by $\overline{\hat{q}_{\alpha,t}^{DQ} - \hat{q}_{\alpha,t}^{opt}}$ where $\hat{q}_{\alpha,t}^{opt}$ is either the VIX-implied quantile, $\hat{q}_{\alpha,t}^{VIX}$, or the IV-implied quantile, $\hat{q}_{\alpha,t}^{IV}$.

¹³We calculate the implied volatility from the Black-Scholes model using the nearest term non-leap option with a positive volume whose strike price is closest to the price of the S&P 500 index at the end of the month.

This approach allows for a volatility risk premium, albeit one that is quite simple. Our methodology can be refined at the cost of having to entertain a model for the volatility risk premium such as the square root specification which have been found to be plagued by its own biases, see Christoffersen et al. (2007). Moreover, by considering deviations from the average historical difference our results reflect differences in the information embedded in the quantile forecasts and option volatility measures, respectively, and so our results are not due to the low average returns associated with investments in call or put options documented by Coval and Shumway (2001).

6.2 Trading Strategies

Our trading strategies focus on options with the two nearest expiration dates in order to straddle the forecast horizon of 30 calendar days used to compute the quantile forecasts. The average time to maturity of the two nearest-term options is generally close to 30 days, thus matching our forecast horizon.

We consider four quantile-based option trading strategies:

1. If

$$\hat{q}_{\alpha,t}^{DQ} - \hat{q}_{\alpha,t}^{opt} > \overline{\hat{q}_{\alpha,t}^{DQ} - \hat{q}_{\alpha,t}^{opt}}, \quad \text{for } \alpha = 0.90 \text{ or } 0.95, \quad (30)$$

so our right-tail quantile forecast exceeds the option-implied quantile forecast by more than their average historical difference, then we buy the call option with the strike price closest to the predicted α -quantile of the index price at the end of month $t + 1$, $\hat{q}_{\alpha,t}$. Otherwise, we do not trade in month t .

2. Conversely, if

$$\hat{q}_{\alpha,t}^{DQ} - \hat{q}_{\alpha,t}^{opt} < \overline{\hat{q}_{\alpha,t}^{DQ} - \hat{q}_{\alpha,t}^{opt}}, \quad \text{for } \alpha = 0.90 \text{ or } 0.95, \quad (31)$$

so the right-tail quantile forecast falls below the option-implied quantile forecast by more than their average historical difference, then we sell the call option with the strike price closest to the predicted α -quantile of the index price at the end of month $t + 1$, $\hat{q}_{\alpha,t}$. Otherwise, we do not trade in month t .

3. If

$$\hat{q}_{\alpha,t}^{DQ} - \hat{q}_{\alpha,t}^{opt} < \overline{\hat{q}_{\alpha,t}^{DQ} - \hat{q}_{\alpha,t}^{opt}}, \quad \text{for } \alpha = 0.05 \text{ or } 0.10, \quad (32)$$

so the left-tail quantile forecast falls below the option-implied quantile forecast by more than their average historical difference, then we buy the put option with the strike price closest to the predicted α -quantile of the index price at the end of month $t + 1$, $\hat{q}_{\alpha,t}$. Otherwise, we do not trade in month t .

4. Conversely, if

$$\hat{q}_{\alpha,t}^{DQ} - \hat{q}_{\alpha,t}^{opt} > \overline{\hat{q}_{\alpha,t}^{DQ} - \hat{q}_{\alpha,t}^{opt}}, \quad \text{for } \alpha = 0.05 \text{ or } 0.10, \quad (33)$$

so the left-tail quantile forecast exceeds the option-implied quantile forecast by more than their average historical difference, then we sell the put option with the strike price closest to the predicted α -quantile of the index price at the end of month $t + 1$, $\hat{q}_{\alpha,t}$. Otherwise, we do not trade in month t .

To gain intuition for these trading rules, note that if (30) is satisfied, then our quantile model predicts a higher chance of a large positive return in the following month than the option-implied quantile. In this situation the option market appears to underpredict the upside potential for the S&P 500, so we buy the matching call option at the current market prices. The intuition for the other strategies is similar and they attempt to take advantage of any discrepancy between the upside or downside potentials of market returns suggested by our quantile forecasts compared with the option-implied quantiles.

To avoid problems associated with stale prices or lack of liquidity, we only trade options that satisfy certain minimum volume constraints. In particular, we only trade options with a volume that is at least 10% of the most traded option on the same day. If there is no such option satisfying the minimum volume constraint, we do not trade in that month.¹⁴

Payoffs from the option strategies are calculated assuming that the investor borrows or lends at the risk-free rate and that any payoff from the exercise of the option is invested at the risk-free rate.¹⁵ For example, payoffs from the strategy in (30) are calculated assuming that the initial purchase is borrowed at the risk-free rate and is paid back when the second option expires and any payoff from the first option is invested at the risk-free rate until the second option expires. Payoffs are calculated similarly for the other strategies and can be written as follows (suppressing, for simplicity, the t - and α -subscripts):

¹⁴There are only few months where we do not trade because of a violation of the minimum volume constraint.

¹⁵We use the continuously compounded 3-month T-bill rate on the last day of each month as the risk-free rate.

- Payoff from Strategy 1 =

$$\begin{cases} -(P_1 + P_2)(1 + r_1^f)^{T_1/360} \\ + \max(S_1 - K_1, 0)(1 + r_2^f)^{T_2/360} \\ + \max(S_2 - K_2, 0), & \text{if (30) holds;} \\ 0, & \text{otherwise.} \end{cases} \quad (34)$$

- Payoff from Strategy 2 =

$$\begin{cases} +(P_1 + P_2)(1 + r_1^f)^{T_1/360} \\ - \max(S_1 - K_1, 0)(1 + r_2^f)^{T_2/360} \\ - \max(S_2 - K_2, 0), & \text{if (31) holds;} \\ 0, & \text{otherwise.} \end{cases}$$

- Payoff from Strategy 3 =

$$\begin{cases} -(P_1 + P_2)(1 + r_1^f)^{T_1/360} \\ + \max(K_1 - S_1, 0)(1 + r_2^f)^{T_2/360} \\ + \max(K_2 - S_2, 0), & \text{if (32) holds;} \\ 0, & \text{otherwise.} \end{cases}$$

- Payoff from Strategy 4 =

$$\begin{cases} +(P_1 + P_2)(1 + r_1^f)^{T_1/360} \\ - \max(K_1 - S_1, 0)(1 + r_2^f)^{T_2/360} \\ - \max(K_2 - S_2, 0), & \text{if (33) holds;} \\ 0, & \text{otherwise.} \end{cases}$$

Here S_1 and S_2 are the prices of the S&P 500 index on the first and second nearest option expiration dates.¹⁶ Similarly, K_1 and K_2 are the strike prices of the options with the first and second nearest expiration dates, respectively, and P_1 and P_2 are the purchase prices of these options. r_1^f is the risk-free rate on the option purchase date, i.e. the last trading day of the month, whereas

¹⁶The expiration date is the Saturday following the third Friday of the expiration month and the exercise-settlement value, S_1 or S_2 , is calculated using the opening price of the S&P 500 Index on the last business day (usually a Friday) before the expiration date. If the stock market does not open on the day on which the exercise-settlement value is determined, then we use the closing price of the S&P 500 Index on the last business day before the expiration date.

r_2^f is the risk-free rate on the last business day before the first option expires.¹⁷ Finally, T_1 is the number of days between the original option purchase date and the expiration date of the second option while T_2 is the number of days between the expiration dates of the two options.

To analyze whether our quantile forecasts provide economically valuable information for option trading, we compare the payoffs from our trading strategies to payoffs from benchmark strategies such as always buying options with matching strikes. Strategy 1 of selectively buying call options is thus compared to always buying call options; strategy 2 of selectively selling call options is compared to always selling call options; strategy 3 of selectively buying put options is compared to always buying put options; finally, strategy 4 of selectively selling put options is compared to always selling put options. Payoffs from these benchmark strategies are computed in a similar fashion to those from the corresponding strategies 1-4. We also do not trade when the minimum volume constraint is not satisfied in which case the benchmark payoff is assumed to be zero.

6.3 Stochastic Dominance Tests

Economic valuation of these trading strategies is made difficult by the nonlinear payoffs on the underlying options. To deal with this issue, we consider whether the payoffs from our selective option trading strategy second order stochastically dominate those from the benchmark based only on market information embedded in the corresponding call or put option prices and the implied volatility estimates. For assets with nonlinear payoffs such as options, the mean and variance are incomplete measures of the return distribution and stochastic dominance measures are more appropriate.

Second order stochastic dominance allows a comparison of payoffs for broader classes of utility than comparisons based on specific functional forms such as power utility and has been used to characterize risk in recent studies such as Post and Levy (2005). If payoffs from the quantile-based option trades second order stochastically dominate benchmark payoffs, then any non-satiated, risk averse option investor should be willing to incorporate information from time-varying quantile predictions into his investment strategy.

To test if the payoffs from the quantile strategies second order stochastically dominate those from the corresponding benchmarks, we use the stochastic dominance tests recently proposed by Linton

¹⁷If the risk-free rate is not available on the purchase date or the last trading day before the first option expires, then we use the first available observation before the corresponding day.

et al. (2005). Specifically, let $\bar{F}_{iN}(y) = \frac{1}{N} \sum_{j=1}^N 1(Y_{ij} \leq y)$ be the empirical cumulative distribution function (CDF) of a random variable Y_i based on a sample with N observations, $\{y_{i1}, \dots, y_{iN}\}$, where y is defined over a grid between $\underline{\mathcal{Y}} = \lfloor \min\{y_{i1}, \dots, y_{iN}\} \rfloor$ and $\bar{\mathcal{Y}} = \lceil \max\{y_{i1}, \dots, y_{iN}\} \rceil$ with increments of $(\bar{\mathcal{Y}} - \underline{\mathcal{Y}})/N$, i.e. $y \in \mathcal{Y} = \{\underline{\mathcal{Y}}, \underline{\mathcal{Y}} + (\bar{\mathcal{Y}} - \underline{\mathcal{Y}})/N, \dots, (\bar{\mathcal{Y}} - \underline{\mathcal{Y}})(N-1)/N, \bar{\mathcal{Y}}\}$. Here $\lfloor \cdot \rfloor$ is the largest smaller integer operator whereas $\lceil \cdot \rceil$ is the smallest larger integer operator.

To test if the payoffs from the quantile-based trading strategy, $Y_{Q,opt}$, second order stochastically dominate those from the associated benchmark, Y_{Bmk} , define the test statistic d_2^* :

$$d_2^* = \max_{y \in \mathcal{Y}} \sqrt{N} [\bar{D}_{Q,opt}^{(2)}(y) - \bar{D}_{Bmk}^{(2)}(y)] \quad (35)$$

where $\bar{D}_i^{(1)}$ is the CDF of y_i while $\bar{D}_i^{(2)}$ is the integrated CDF defined as

$$\begin{aligned} \bar{D}_i^{(1)}(y) &= \bar{F}_{iN}(y) \\ \bar{D}_i^{(2)}(y) &= \int_{-\infty}^y \bar{D}_i^{(1)}(z) dz. \end{aligned} \quad (36)$$

Zero values of d_2^* suggest that the integrated CDF of the quantile trading rule uniformly falls below the integrated CDF of the benchmark. This makes the quantile trading rule desirable for non-satiated investors with concave utility. The null hypothesis that payoffs from the quantile trading rule, $Y_{Q,opt}$, second order stochastically dominate those from the benchmark, Y_{Bmk} , is tested against the alternative that the benchmark dominates the quantile trading strategy:

$$\begin{aligned} \mathbf{H}_0 : d_2^* &\leq 0 \\ \mathbf{H}_1 : d_2^* &> 0. \end{aligned} \quad (37)$$

p -values of this test statistic can be calculated using the subsampling methods proposed by Linton et al. (2005). Let $W_j = (Y_{Q,opt,j}, Y_{Bmk,j})$ denote the j -th observation of the paired quantile and benchmark payoffs and define the subsample with b consecutive observations as $\{W_j, W_{j+1}, \dots, W_{j+b-1}\}$ for $j = 1, \dots, N-b+1$. Subsamples are drawn from the original data without replacement. Defining the test statistic for subsample j as $d_{2,j}^*$, the p -value for the null hypothesis is given by

$$p = 1 - \frac{1}{N-b+1} \sum_{j=1}^{N-b+1} \mathbf{1}\{d_{2,j}^* > 0\}, \quad (38)$$

where $\mathbf{1}\{\cdot\}$ is the indicator function. Further details of the subsampling approach can be found in Linton et al. (2005) who also suggest methods to choose the best subsample size.

Using these p -values, we can test whether the payoffs from the quantile-based option trading strategies second order stochastically dominate the benchmark. Suppose we fail to reject the null hypothesis that the payoffs from our strategies stochastically dominate the benchmark while we conversely *do* reject that the benchmark payoffs stochastically dominate those from the quantile-based strategy. Then we conclude that our quantile strategy is ‘better’ than the benchmark and that our quantile forecasts provide useful information for option trading.

6.4 Empirical Results

Since our data on S&P 500 options begin in January 1990, our first trade uses the quantile forecasts for February 1990 and the options prices from the last trading day of January 1990. The last trade is for December 2005. This results in 191 monthly payoffs which we use to test whether the payoffs from our quantile strategies stochastically dominate those from the benchmarks.¹⁸

Table 8 presents empirical results from our analysis. First consider the results based on the VIX which are shown in panel A. Irrespective of which of the upper quantiles is considered ($\alpha = 0.90$ or $\alpha = 0.95$), the trading strategies based on the call options strongly suggest that there is important economic information in the dynamic quantile forecasts. In particular, we always fail to reject the null that the payoffs based on the quantile forecasts second order stochastically dominate the benchmark payoffs, while we can reject the converse proposition, i.e. that the payoffs from the benchmark strategy dominate those from the quantile strategy.¹⁹ These findings are consistent with the earlier evidence that the dynamic quantile forecasts perform well in the upper tail of the return distribution.

Turning to the put options and thus forecasts of events in the left tail of the return distribution, it appears that the payoffs from the quantile-based strategy of selectively buying put options second order stochastically dominate those from the benchmark. The evidence is inconclusive, however, for the fourth strategy that selectively sells put options. This is consistent with some evidence of time-varying predictability of quantiles in the lower parts of the return distribution although the

¹⁸In our empirical analysis, we set the subsample size, b , to 50 observations which gives a total of 142 subsamples to approximate the distribution of the test statistic. We also tried using a subsample size of 56 which corresponds to four times the square root of the sample size, one of the approaches suggested by Linton et al. (2005). The results are very similar to the ones reported here.

¹⁹A test statistic of zero means that the integrated CDF of the payoffs associated with the dynamic quantile forecasts at every point fall below the integrated CDF of the benchmark.

evidence appears to be somewhat weaker compared to that for the upper quantiles.

Very similar results are obtained when the payoffs associated with the dynamic quantile forecasts are compared to those from the Black-Scholes implied quantiles as shown in Panel B of Table 8. We find that the dynamic quantile-based trades stochastically dominate the benchmark payoffs for all of the experiments involving call options. Moreover, the strategies of selectively buying put options stochastically dominate its passive benchmark while the results are inconclusive for the strategy of selectively selling put options.

7 Conclusion

We use dynamic quantile models to explore the extent to which different parts of the distribution of stock returns are predictable by means of economic state variables. Consistent with earlier studies we find little evidence to suggest that the center of the return distribution can be predicted. However, our findings also suggest that many of the predictor variables proposed in the finance literature, including the T-bill rate, inflation, the default yield, stock variance, payout ratio and net equity issues contain valuable information for predicting parts of the return distribution. Our finding that many state variables work either in both tails but not in the center or in one tail but not in both suggests that variations in the conditional quantiles of the return distribution are not simply due to time-varying volatility. Interestingly, the evidence in support of out-of-sample predictability of stock returns is strongest in the right tail of the return distribution. While most previous work has focused on ‘downside risk’, the possibility of predicting periods with strong upside potential has not received nearly as much attention.

Our findings that predictability of return quantiles can be used to improve portfolio allocations for risk averse investors or to trade call and (to some extent) put options with desirable payoff distributions suggest promising economic gains from using information on the full return distribution. This could prove important also to studying hedge fund returns which are known to have option-like return characteristics (Mitchell and Pulvino (2001)), structured products (Ang et al. (2005)) and other types of investments. It is our hope that the results in this paper will give rise to further investigation of predictability of the distribution of stock returns.

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Table 1: Predictor Variables

Variable	Description	Sample
d/p	Dividend Price Ratio	02/1871-12/2005
d/y	Dividend Yield	02/1871-12/2005
e/p	Earnings Price Ratio	02/1871-12/2005
e10/p	Smoothed Earnings Price Ratio	12/1880-12/2005
b/m	Book to Market Ratio	03/1921-12/2005
tbl	T-bill	02/1920-12/2005
lty	Long Term Yield	01/1919-12/2005
tms	Term Spread	02/1920-12/2005
dfy	Default Yield Spread	01/1919-12/2005
dfr	Default Return Spread	01/1926-12/2005
csp	Cross Sectional Premium	05/1937-12/2002
ltr	Long Term Rate of Return	01/1926-12/2005
svar	Stock Variance	02/1885-12/2005
d/e	Dividend Payout Ratio	02/1871-12/2005
ntis	Net Equity Expansion	12/1926-12/2005
infl	Inflation	02/1913-12/2005

Note: This table lists the 16 predictor variables used in the study. The data source is Goyal and Welch (2007).

Table 2: OLS Estimates of Regression Coefficients for Univariate Return Prediction Models

Predictor Variable	Estimate
Dividend Price Ratio	0.0000
Dividend Yield	0.0017
Earnings Price Ratio	0.0051
Smoothed Earnings Price Ratio	0.1332
Book to Market Ratio	0.0106
T-bill Rate	-0.0890
Long Term Yield	-0.0604
Term Spread	0.2040
Default Yield Spread	0.0648
Default Return Spread	0.1436
Cross Sectional Premium	1.7783**
Long Term Rate of Return	0.0935
Stock Variance	-0.1533
Dividend Payout Ratio	-0.0086
Net Equity Expansion	-0.2177***
Inflation	-0.4757*

Note: These coefficient estimates are based on full-sample OLS regressions of monthly excess returns on the S&P 500 index against each of the predictor variables listed in the rows. Data samples are listed in Table 1.

* indicates significance at the 10% level

** indicates significance at the 5% level

*** indicates significance at the 1% level

Table 3: Coefficient Estimates for Linear Quantile Prediction Models

	Quantile												Bonferonni
	0.05	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	0.95	Bonferonni	
d/p	-0.6960	-0.3134	-0.2447	0.0428	-0.2313	-0.4543	-0.3206	-0.0446	-0.0389	0.0589	0.6702	1.0000	
d/y	-0.3162	0.1588	-0.1525	0.1244	-0.0668	-0.4070	-0.2592	0.0280	-0.0392	0.0558	0.1216	1.0000	
e/p	1.3633**	0.9623**	-0.0034	0.2795	0.4183	-0.0226	0.4531	0.6810*	0.2216	-0.0373	-0.2612	0.2530	
e10/p	-0.3834	-0.1124	-0.0581	0.0143	0.0311	0.0037	0.1152	0.1611***	0.1591*	0.2811*	0.4193*	0.0330	
b/m	-4.9035**	-2.2920	-1.0497	-0.8978	-0.7111	-0.6072	0.5946	1.0959	1.6289***	1.9554	4.9288***	0.0330	
tbl	0.1147	0.0142	-0.0430	-0.1038	-0.1468***	-0.1390**	-0.1349**	-0.1361	-0.1298**	-0.1920***	-0.2384***	0.0000	
lty	0.1841	0.0995	0.0062	-0.0870	-0.1221**	-0.1138	-0.1300*	-0.1233	-0.1176**	-0.1804**	-0.0990	0.1980	
tms	-0.0589	0.1501	0.3250**	0.1257	0.2009*	0.1262	0.0908	0.0939	0.1738	0.5073**	0.7515**	0.2530	
dŷ	-4.1446***	-2.9432***	-1.4745**	-0.9104**	-0.4549	0.0387	0.5415*	0.6163*	1.2677***	2.1080***	3.4645***	0.0000	
dfr	0.7128*	0.1736	0.2696	0.1163	0.0865	0.0931	-0.0787	0.0825	0.1208	0.2046	0.1780	0.9570	
csp	3.8071**	2.3905*	0.4890	2.1476**	2.0656**	1.2664	1.4879	2.2975***	1.6341**	2.0216*	1.4020	0.0990	
ltr	-0.1741	0.0792	0.0342	-0.0234	0.0122	0.0472	0.1409**	0.0980	0.0682	0.1800	0.2504	0.2310	
svar	-10.2609***	-4.1415**	-2.6219***	-2.2660***	-1.2624**	-0.6413	0.2569	2.1804**	2.7451***	6.2257***	8.5168***	0.0000	
d/e	-0.0411***	-0.0275***	-0.0086	-0.0073	-0.0086**	-0.0144***	-0.0136***	-0.0142***	-0.0038	0.0042	0.0195	0.0000	
ntis	-0.4655***	-0.4646**	-0.3404***	-0.1712*	-0.1527	-0.1108	-0.0756	-0.0676	-0.0338	-0.0165	0.0053	0.0220	
infl	0.3120	-0.3674	-0.4931*	-1.0195***	-1.0710***	-0.9347***	-0.9260***	-0.6739**	-0.6043**	-0.8240	-1.1059*	0.0000	

Note: For each quantile $\alpha = \{0.05, 0.10, \dots, 0.90, 0.95\}$ the table reports the slope coefficients obtained from quasi-maximum likelihood estimation of univariate quantile models using the samples listed in Table 1. In each case the dependent variable is the excess return on the S&P 500 index. The significance of the coefficient estimates is based on bootstrapped standard errors. The final column lists Bonferroni p-values for a joint test across all quantiles that the slope coefficients in the quantile model are equal to zero. The coefficient estimates of d/p, d/y, e/p and b/m have been multiplied by 100.

* indicates significance at the 10% level

** indicates significance at the 5% level

*** indicates significance at the 1% level

Table 4: Out-of-sample Coverage Probabilities

	Quantile				
	0.05	0.1	0.5	0.9	0.95
d/p	0.0532	0.0903	0.5046	0.8843	0.9444
d/y	0.0486	0.0926	0.5000	0.8843	0.9444
e/p	0.0463	0.0949	0.4861	0.8912	0.9514
e10/p	0.0486	0.0903	0.4769	0.8866	0.9421
b/m	0.0625	0.1134	0.5278	0.8912	0.9375
tbl	0.0417	0.0787	0.4653	0.8681	0.9560
lty	0.0394	0.0741	0.4491	0.8704	0.9630
tms	0.0741	0.1204	0.5347	0.9167	0.9583
dfy	0.0486	0.0949	0.5023	0.9074	0.9560
dfr	0.0463	0.0833	0.5139	0.9167	0.9583
csp	0.0505	0.0960	0.4848	0.8813	0.9394
ltr	0.0718	0.1273	0.5417	0.9306	0.9676
svar	0.0486	0.1157	0.4907	0.9028	0.9560
d/e	0.0486	0.1134	0.5162	0.9097	0.9514
ntis	0.0579	0.1227	0.5347	0.9144	0.9583
infl	0.0486	0.0926	0.4907	0.9051	0.9537
EW Combination	0.0486	0.0972	0.5023	0.8958	0.9560
GARCH (1,1)	0.0625	0.0949	0.4606	0.9259	0.9606
Prevailing Quantile	0.0532	0.0995	0.4630	0.8843	0.9398

Note: This table reports the proportion of actual stock returns in the out-of-sample period (1970:01 - 2005:12) that fall below the predicted quantile. If the model is correctly specified this proportion should be equal to the coverage probability listed at the top of each column. Rows 1-16 report results for the dynamic quantile specification based on the predictor variables listed in each row. The row 'EW Combination' reports results for the equal-weighted combination that averages the dynamic quantile forecasts across the individual models. The GARCH(1,1) model accounts for time-varying volatility, while the prevailing quantile model assumes a constant return distribution. For all models, the parameters are estimated recursively without making use of full-sample information.

Table 5: Performance of Quantile Forecasts: Mean Out-of-sample Loss

	Quantile				
	0.05	0.1	0.5	0.9	0.95
d/p	5.5325	8.6594	16.9466	7.1562	4.3128
d/y	5.5377	8.6755	16.9264	7.1469	4.2874
e/p	5.5265	8.6601	16.9785	7.1392	4.2612
e10/p	5.5501	8.6841	17.0794	7.1194	4.2700
b/m	5.8555	9.1355	17.1374	7.2975	4.2745
tbl	5.4715	8.7043	16.9280	7.2693	4.4277
lty	5.5751	8.8132	16.9988	7.4253	4.4752
tms	5.3816	8.6087	17.0521	7.1905	4.3665
dfy	5.6638	8.8022	17.0254	7.0867	4.3031
dfr	5.4113	8.6974	17.0025	7.1913	4.3750
csp	5.6114	8.8374	17.5161	7.3926	4.5084
ltr	5.4742	8.6973	17.1170	7.2641	4.3513
svar	5.4263	8.5910	16.9599	7.0681	4.3282
d/e	5.5314	8.6965	17.0952	7.0684	4.3216
ntis	5.7265	8.8886	17.0437	7.1779	4.3492
infl	5.5254	8.6535	16.7376	7.1001	4.3418
EW Combination	5.4690	8.5754	16.8877	7.0601	4.3012
GARCH(1,1)	5.5293	8.6972	16.9674	7.1238	4.3243
Prevailing Quantile	5.4560	8.6685	16.9318	7.2852	4.4085
Number of models					
better than PQ	3	5	3	13	13

Note: This table reports the average loss under the tick loss function computed over the out-of-sample period from 1970:1 to 2005:12. An expanding window of data is used to estimate the parameters of the forecasting models which are updated at each point in time using only historically available data. EW Combination refers to quantile forecasts from the equal-weighted combination of the 16 univariate dynamic quantile forecasts. Quantiles from the GARCH(1,1) model account for time-varying volatility while the prevailing quantile (PQ) estimates assume that the quantiles are constant over time. Boldfaced numbers indicate the best model for each quantile.

Table 6: Combination Weights for the Quantile Prediction and Prevailing Quantile Models

Predictor	0.05		0.1		0.5		0.9		0.95	
Variable	DQ	PQ	DQ	PQ	DQ	PQ	DQ	PQ	DQ	PQ
d/p	0.5007	0.3400	-0.0928	0.9298	-0.1251	7.2296	0.8636***	0.2606	0.7146***	0.3254**
d/y	0.3380**	0.4787***	-0.0716	0.9073**	-0.1696	7.2967*	0.8520***	0.2657	0.6494***	0.3890***
e/p	0.4611***	0.3106*	-0.0906	0.9223**	0.7561	6.2341***	0.9180***	0.1868	0.4746**	0.5450**
e10/p	0.4536	0.4169	-0.0383	0.8592	-1.7106	7.7517***	1.0221***	0.1455	0.6595***	0.3883***
b/m	0.4298***	0.4219***	-0.0790	0.9078***	-1.9496***	17.1997***	0.8494*	0.2588	0.6268***	0.4277***
tbl	0.3368***	0.3983***	0.1981*	0.5478***	0.2332	7.3252***	0.7038*	0.4423	0.5885***	0.3841*
lty	0.2137	0.5914**	0.0181	0.7915*	0.1246	8.4382***	0.5651**	0.6420***	0.6686**	0.2846
tms	0.3707***	0.4298***	0.1637	0.6777***	0.7863	3.8384	0.6984	0.3117	0.6097***	0.3456*
dfy	-0.0965	0.9246***	-0.1008	0.9259***	-1.0545	10.5189*	1.2145***	-0.1662	0.6516**	0.3394
dfz	0.1958	0.5379**	-0.0049	0.8112	-0.2625	7.7075	0.6649***	0.3598	0.5807***	0.4135***
csp	0.1766	0.5287	-0.0528	0.8772	0.2405	8.2845***	1.1190***	0.0545	0.6606**	0.3831
ltr	0.4379**	0.4522***	0.2678	0.5966***	0.4003	4.1356	0.6716***	0.3294	0.5762*	0.3477
svar	0.2959	0.5745	0.0926	0.7491	0.9182	4.5338	1.1854***	-0.0955	0.4315***	0.5581***
d/e	0.5371	0.2911	-0.0992	0.9227***	-0.3898	8.7521***	0.9633***	0.0965	0.6663**	0.3515
ntis	-0.1387	0.9819**	-0.3846	1.2671***	0.7388	3.1927	1.2891***	-0.2807	0.5966***	0.3496*
infl	0.5235***	0.2710***	-0.0347	0.8401***	3.2092***	-2.3225	1.1837***	-0.1461	0.6518***	0.3425
EW	0.5134***	0.2918**	-0.0404	0.8619	-0.1065	7.1129	1.1819***	-0.1054	0.6663*	0.3261

Note: This table reports the estimated combination weights on the out-of-sample forecasts from 1970-2005 generated by the univariate dynamic quantile (DQ) specifications and the prevailing quantile (PQ) model which assumes a constant return distribution. Significant weights on the DQ forecasts indicate that they help improve on the forecasting performance of the PQ model. The final row lists the combination weights on the equal-weighted average of the individual dynamic quantile forecasts.

* indicates significance at the 10% level

** indicates significance at the 5% level

*** indicates significance at the 1% level

Table 7: Economic Significance Results

	Certainty Equivalent Return		
	Log Utility	$\gamma=2.5$	$\gamma=5.0$
d/p	6.36%	5.75%	5.68%
d/y	6.32%	5.61%	5.75%
e/p	6.62%	5.55%	5.64%
e10/p	6.10%	5.63%	5.73%
b/m	5.34%	4.42%	2.83%
tbl	7.82%	7.29%	6.64%
lty	7.53%	6.90%	6.46%
tms	9.48%	8.05%	6.33%
dfy	5.89%	4.35%	4.83%
dfr	5.99%	4.74%	5.05%
csp	6.74%	6.52%	6.45%
ltr	8.35%	7.65%	6.08%
svar	6.29%	5.64%	5.68%
d/e	6.69%	4.53%	4.24%
ntis	6.68%	5.27%	4.89%
infl	7.87%	7.28%	6.69%
EW combination	8.47%	6.92%	6.47%
GARCH (1,1)	6.44%	6.14%	6.07%
PQ	6.52%	6.20%	6.09%

Note: This table reports the certainty equivalent return for an investor with power utility and coefficient of relative risk aversion, γ . Each month during the period 1970-2005, the investor uses out-of-sample forecasts of return quantiles to form a portfolio of stocks (tracked by the S&P500 index) and T-bills. The state variables used to form quantile predictions are listed in the individual rows. EW combination is the equal-weighted quantile forecast, GARCH(1,1) produces forecasts of stock returns from a Generalized ARCH model, while the prevailing quantile (PQ) model assumes a constant return distribution but updates its parameters as new data arrives.

Table 8: Second Order Stochastic Dominance Tests Based on Option Trades

(a) VIX-implied Quantile

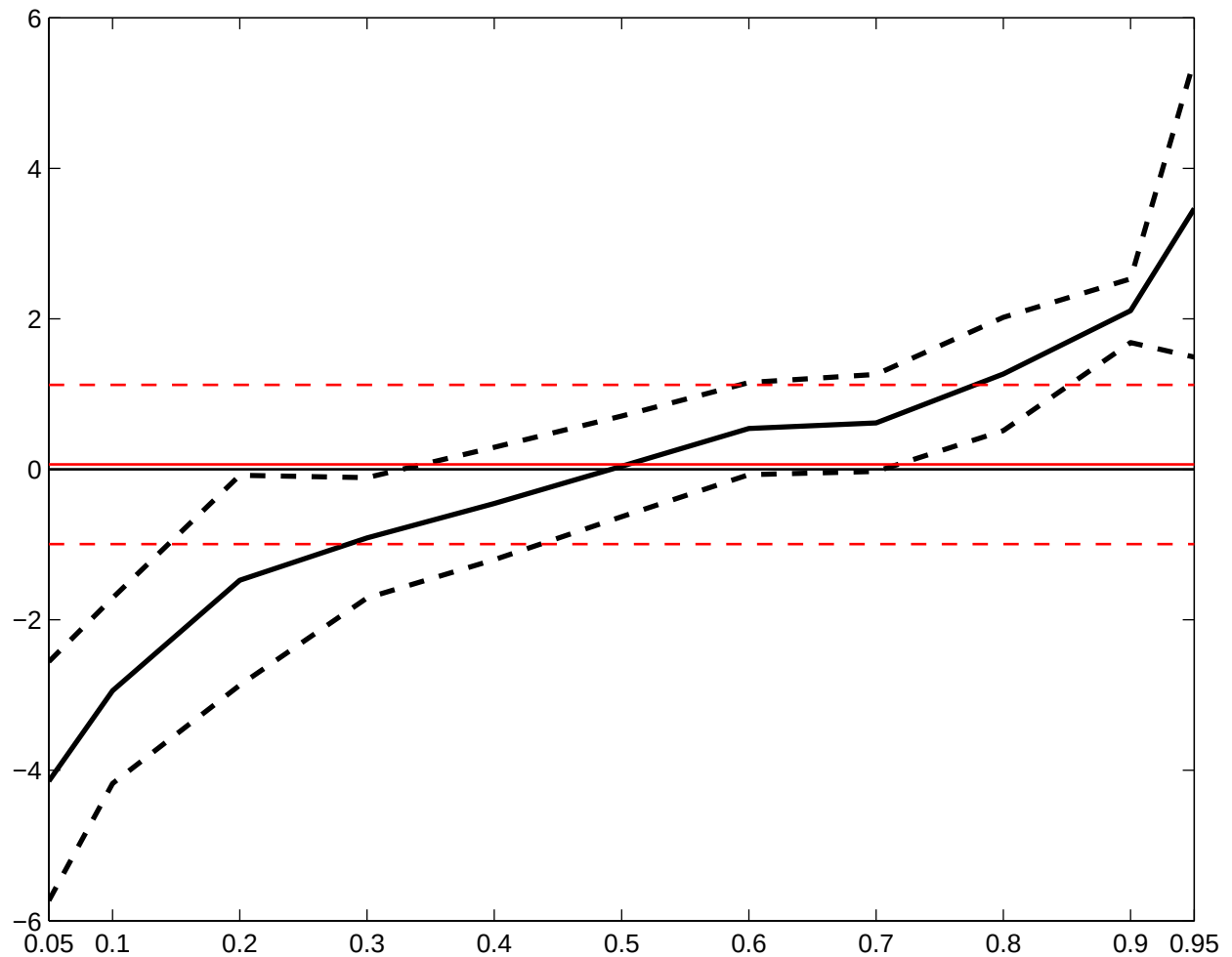
	Test Statistic for H ₀ : Active Strategy SSD Benchmark Strategy	Test Statistic for H ₀ : Benchmark Strategy SSD Active Strategy
Strategy 1 for $\alpha = 0.90$	0.0000	84.5860***
Strategy 1 for $\alpha = 0.95$	0.0000	56.3660***
Strategy 2 for $\alpha = 0.90$	12.9520	28.0750*
Strategy 2 for $\alpha = 0.95$	23.2270	29.3770**
Strategy 3 for $\alpha = 0.05$	0.0000	78.6530***
Strategy 3 for $\alpha = 0.10$	0.0000	84.0790***
Strategy 4 for $\alpha = 0.05$	24.6020	18.7410
Strategy 4 for $\alpha = 0.10$	30.5350	13.3860

(b) Black-Scholes-implied Quantile

	Test Statistic for H ₀ : Active Strategy SSD Benchmark Strategy	Test Statistic for H ₀ : Benchmark Strategy SSD Active Strategy
Strategy 1 for $\alpha = 0.90$	0.0000	91.5320***
Strategy 1 for $\alpha = 0.95$	0.0000	63.9640***
Strategy 2 for $\alpha = 0.90$	0.0000	36.9020***
Strategy 2 for $\alpha = 0.95$	6.8020	38.6390***
Strategy 3 for $\alpha = 0.05$	0.0000	77.8570***
Strategy 3 for $\alpha = 0.10$	0.0000	79.7380***
Strategy 4 for $\alpha = 0.05$	26.6280	17.4380
Strategy 4 for $\alpha = 0.10$	25.4700	22.3580

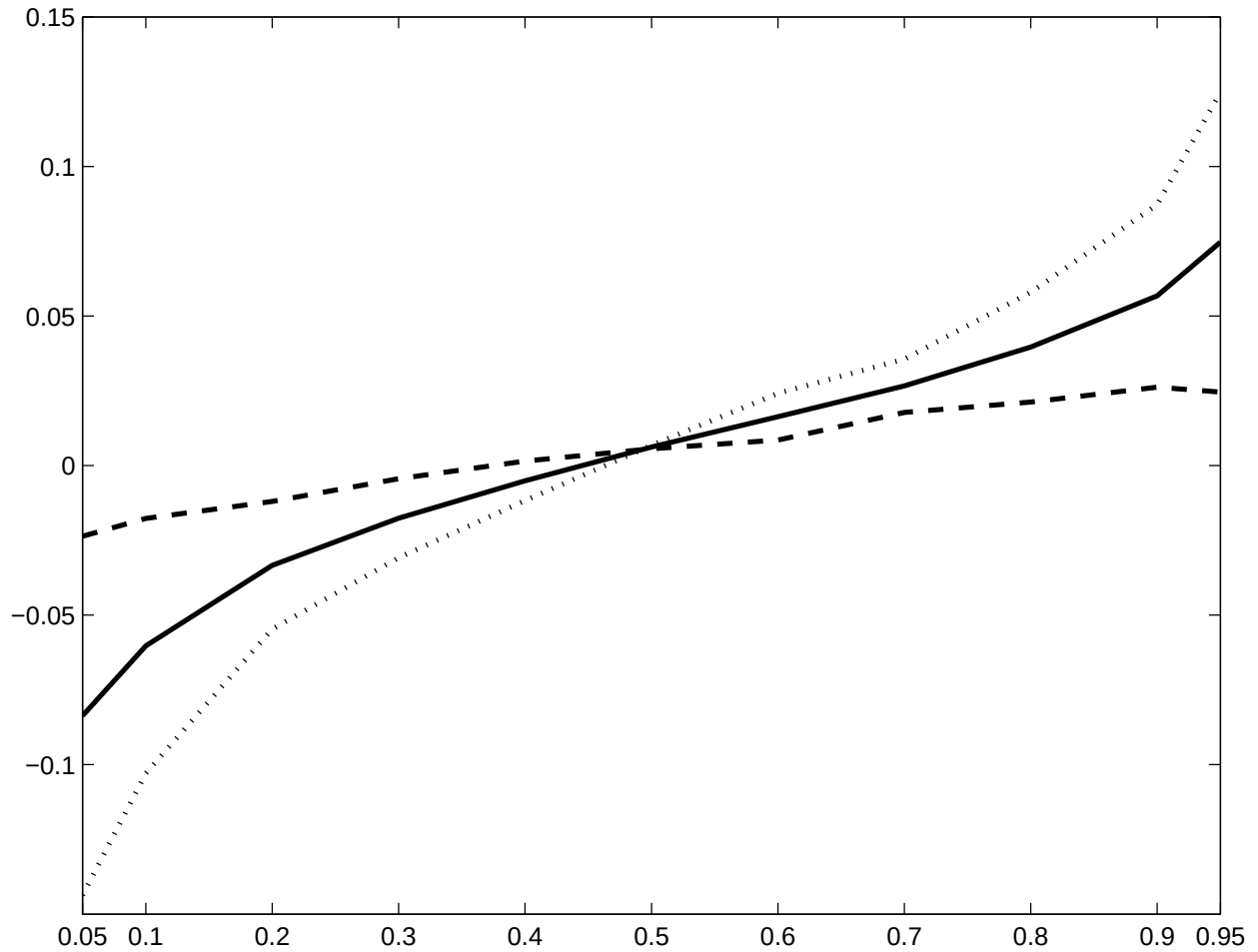
Note: This table reports the outcome of the Linton et. al (2005) tests for second order stochastic dominance applied to a pair of payoff distributions. The first set of payoffs come from selectively trading options based on a comparison of equal-weighted dynamic quantile forecasts with quantile forecasts implied by an assumption that stock returns are normally distributed with constant mean and volatility given by the VIX (Panel (A)) or the Black-Scholes implied volatility (Panel (B)). The second set of payoffs arise from benchmark strategies of always buying or selling call or put options. Strategy 1 buys call options if the dynamic quantile forecasts in the right tail ($\alpha = 0.90$ or $\alpha = 0.95$) exceed the option-implied quantiles by more than their historical margin. The benchmark for this case is to always buy call options. Strategy 2 sells call options if the dynamic quantile forecasts in the right tail fall below the option-implied quantiles. The benchmark for this case is to always sell call options. Strategy 3 buys put options if the dynamic quantile forecasts in the left tail ($\alpha = 0.05$ or $\alpha = 0.10$) fall below the option-implied quantiles by more than their historical margin. The benchmark for this case is to always buy put options. Strategy 4 sells put options if the dynamic quantile forecasts in the right tail exceed the option-implied quantiles. The benchmark for this case is to always sell put options.

Figure 1: Slope Coefficients from the Linear Quantile Model and OLS Estimation



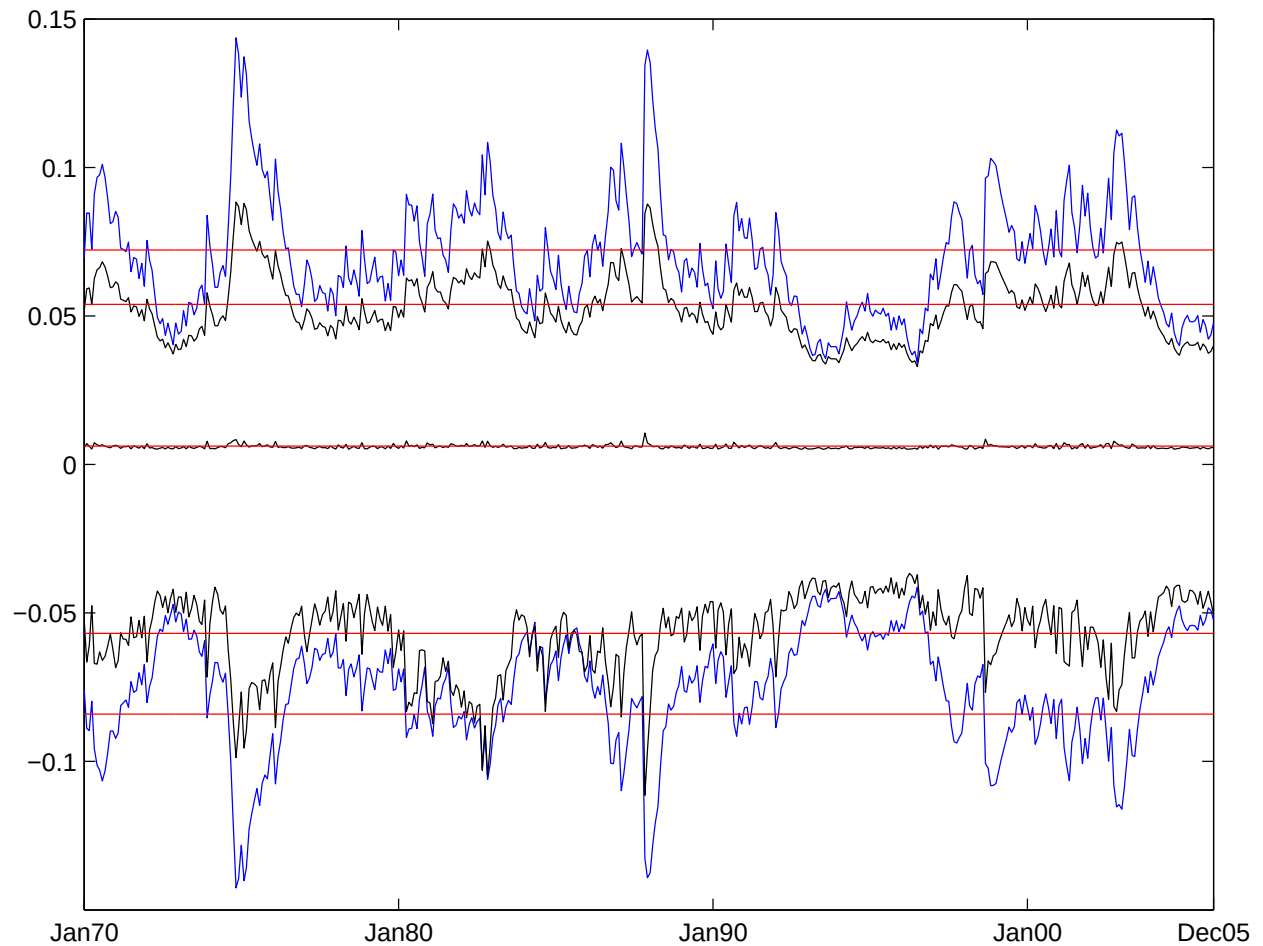
Note: This figure plots the slope coefficients from the quantile model that includes the default yield as a predictor variable (black solid line) and the 95% confidence intervals based on bootstrapped standard errors (black dashed line) along with the corresponding OLS slope coefficient (red solid line) and the OLS 95% confidence intervals based on HAC standard errors (red dashed line).

Figure 2: Quantile Function Using the Default Yield as a Predictor Variable



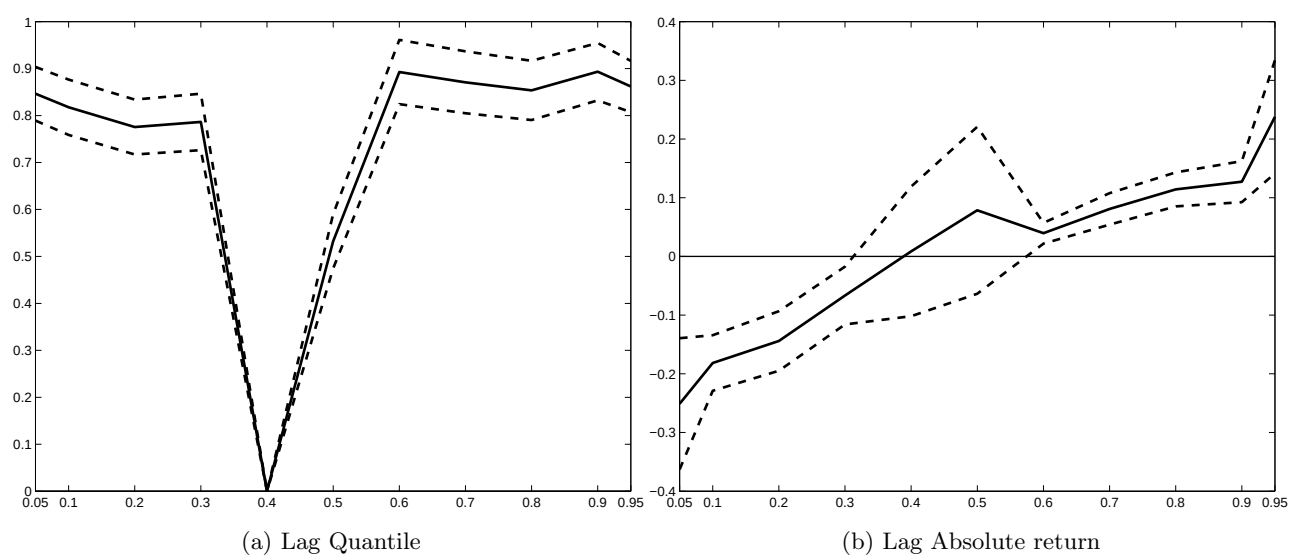
Note: This figure plots the quantile function of returns using the default yield as a predictor variable. The solid line sets the predictor variable to its sample mean; the dotted line sets the predictor variable at its mean plus two standard deviations; the dashed line sets the predictor variable at its mean minus two standard deviations.

Figure 3: Time Series of Quantile Forecasts



Note: This figure plots the 5% (bottom blue line), 10% (bottom black line), 50% (middle black line), 90% (top black line) and 95% (top blue line) conditional quantiles using estimates of the dynamic quantile model with the default yield as a predictor variable. The horizontal lines plot the corresponding full-sample estimates of the constant quantiles of the return distribution.

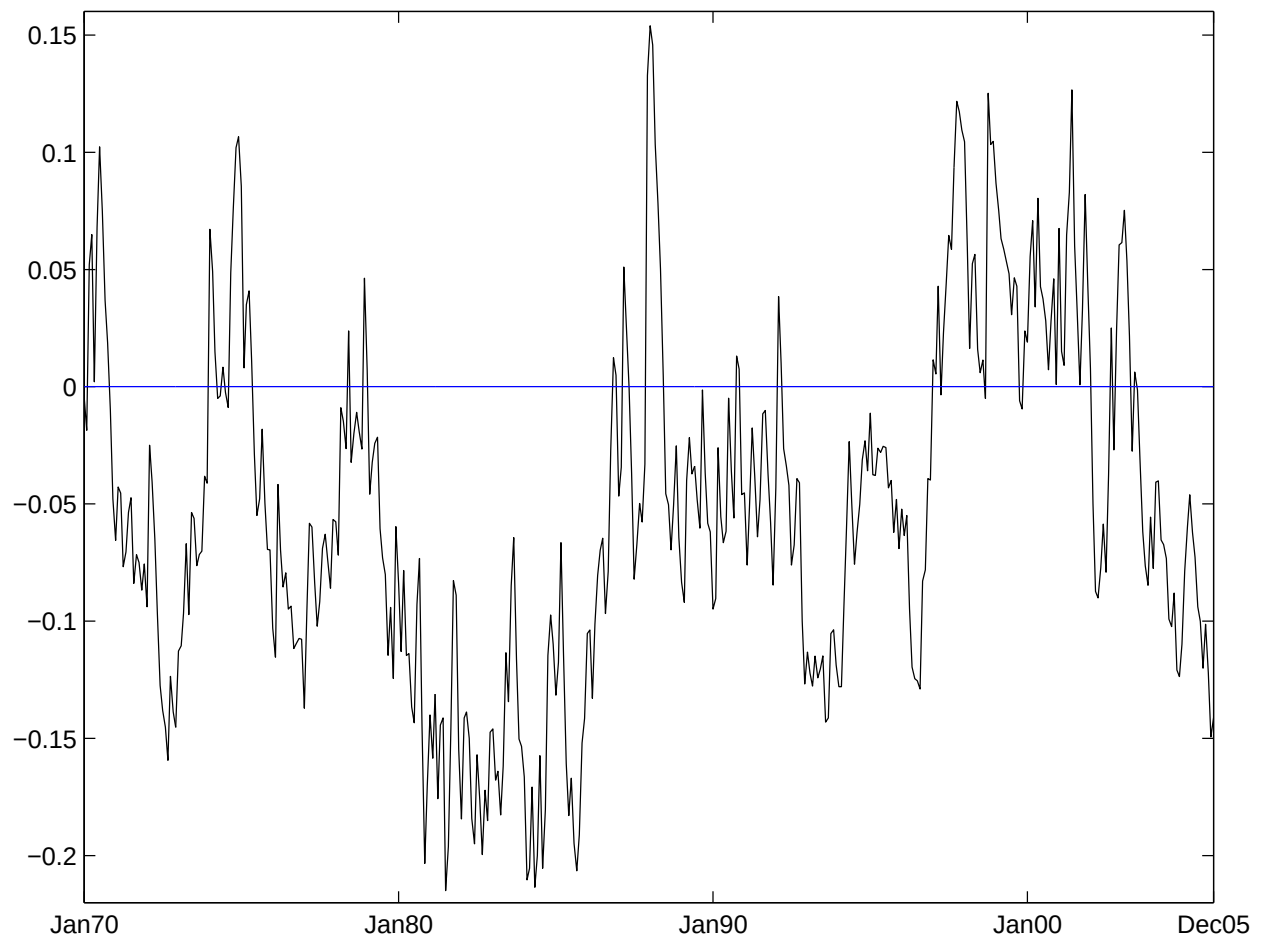
Figure 4: Coefficient Estimates of the Lagged Quantile and Lagged Absolute Returns in the Dynamic Quantile Model



Note: The left window plots the slope coefficients $\beta_{2,\alpha}$ of the lagged conditional quantile while the lower window plots the slope coefficients $\beta_{3,\alpha}$ of the lagged absolute return based on a dynamic quantile model

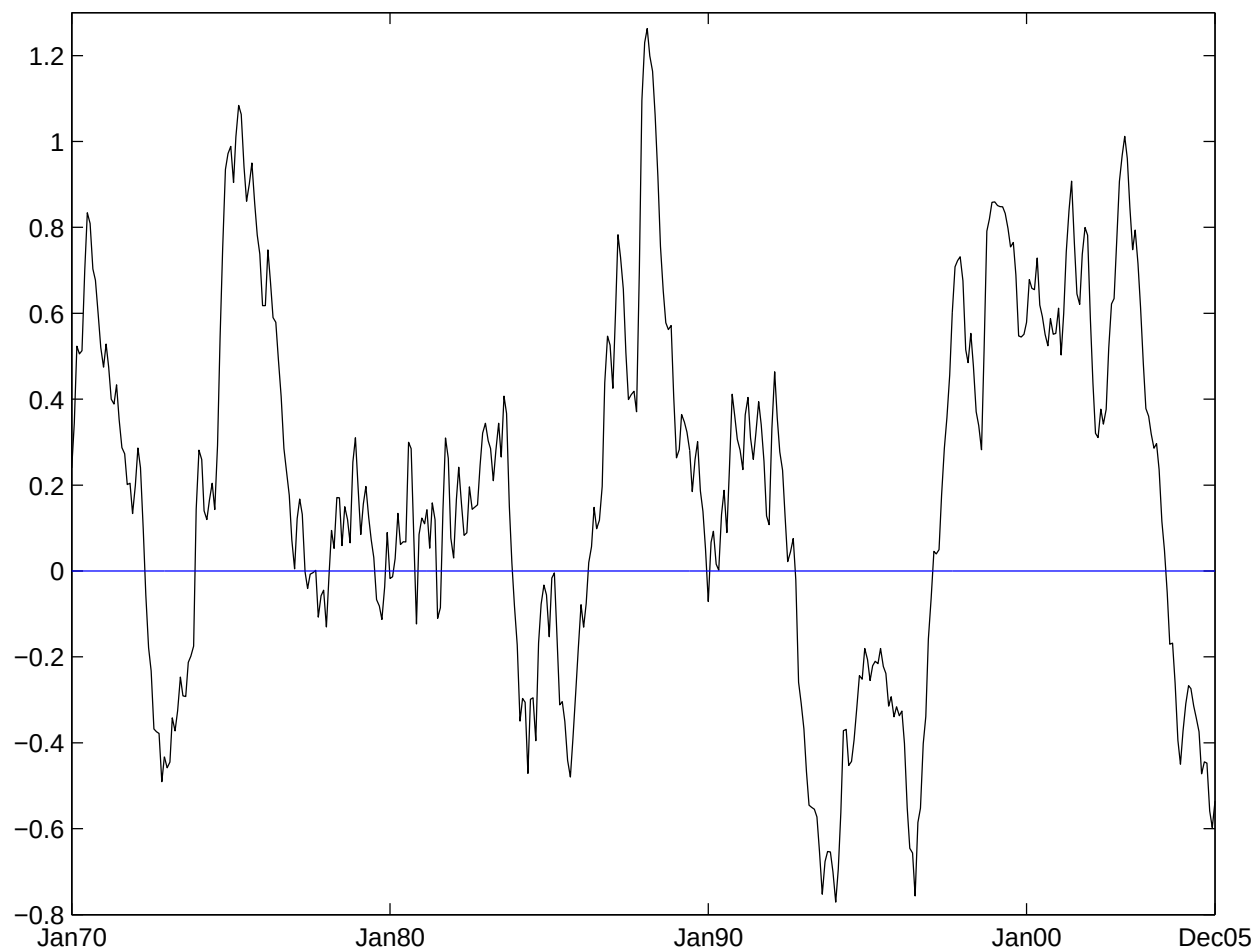
$$q_\alpha(r_{t+1}|\mathcal{F}_t) = \beta_{0,\alpha} + \beta_{1,\alpha}x_t + \beta_{2,\alpha}q_\alpha(r_t|\mathcal{F}_{t-1}) + \beta_{3,\alpha}|r_t|.$$

Figure 5: Conditional Skewness of Returns



Note: This figure plots the conditional skewness of returns based on dynamic quantile estimates that include the default yield as a predictor variable.

Figure 6: Conditional Kurtosis of Returns



Note: This figure plots the conditional excess kurtosis of returns based on dynamic quantile estimates that include the default yield as a predictor variable.